

AA SED MSc + MArch Sustainable Environmental Design 2024-25

Architectural Association School of Architecture

Term 2 Project

Canary Wharf: HSBC Building

The Vertical Village

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September 2025

2.Site Study

2.4- Location

With a latitude of 51.50528°N and a longitude of -0.017389°W, the HSBC building is located at Unit 8, Canada Square, Canary Wharf, London, United Kingdom.

London has an oceanic climate (Cfb) according to the Köppen Climate Classification, characterized by mild summers, cool winters, and no dry season. Rainfall is relatively evenly distributed throughout the year.

Mean Annual Temperature: 11°C
Monthly Precipitation: 36–60 mm
Relative Humidity: 65–84%
Prevailing Wind: Southwest, with speeds ranging from 4 m/s in warmer months to 6 m/s in colder months.

Based on outdoor climate analysis at 8 Canada Square using Meteonorm, three distinct seasons were identified: warm, transitional, and cold.

2.5- Canary Wharf Outdoor Climate

For the purpose of this study, our weather data was obtained from the CBE Clima Tool, Meteonorm and London City Airport Weather Station which is located towards south-east of our site (Figure 2.1.1).

A yearly weather graph was generated using the EPW from the London City Airport and comparing it to RP 8.0 Scenario for the year 2100, it shows that the site experiences an average yearly temperature of 25°C, with the lowest average annual temperature being 21°C and highest average of 29°. Between October and May, the average temperatures are below the comfort band range (ASHRAE 55) except for the period between May and September where the average daily temperatures stay within the range of comfort other than for some days in June where the temperatures rise above. The wind rose from the site illustrates that the wind direction is predominantly from the south or southwest for most of the year with an average wind speed of 5m/s in the summer and 8m/s in the winter. The highest average speed is roughly 9 m/s towards higher levels in the building. (Figure 2.1.2).

To summarise, the following data from the climate is used to inform our study:

Warm Season: Average temperatures exceed the comfort band, peaking at 23–25°C in 2025 and over 28°C by 2100, increasing the risk of overheating. Solar radiation also intensifies, with energy intensity reaching 834 W/m², further contributing to heat stress.

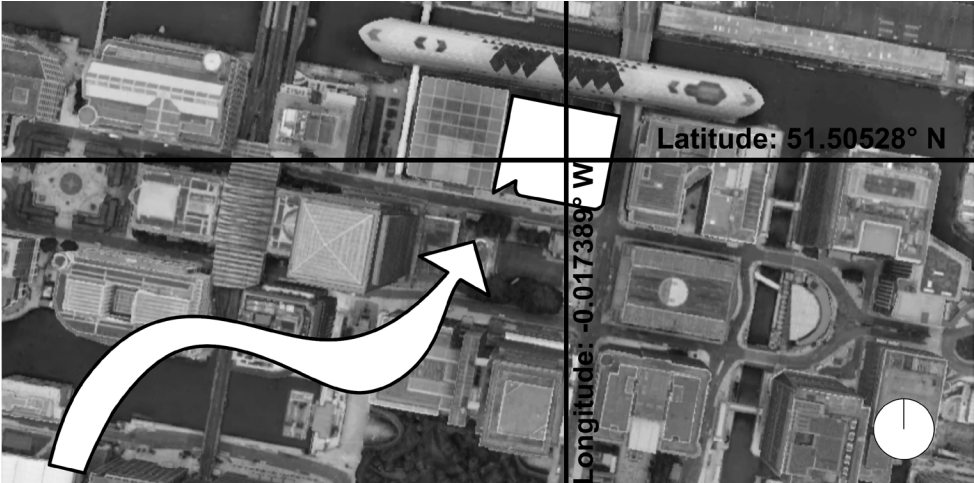
Cold Season: Low temperatures, combined with strong southwesterly winds exceeding 8 m/s, significantly increase heat loss and wind chill effects, reducing outdoor comfort.

Future Projections (2100): A temperature rise of over 5°C is expected, shifting thermal comfort and increasing cooling demands.

Wind Conditions: Prevailing SW winds impact pedestrian comfort and building aerodynamics.

Relative Humidity: Ranges from 65–86%, with higher humidity levels exacerbate thermal discomfort during warm periods.

To improve comfort and energy efficiency, the proposal will explore passive design strategies, including balconies, double-skin facades, shading, ventilation optimization, wind mitigation, and operable windows.



Strong southwest winds exceed 8 m/s
Figure 2.5.1- Prevailant Winds- Source- Author

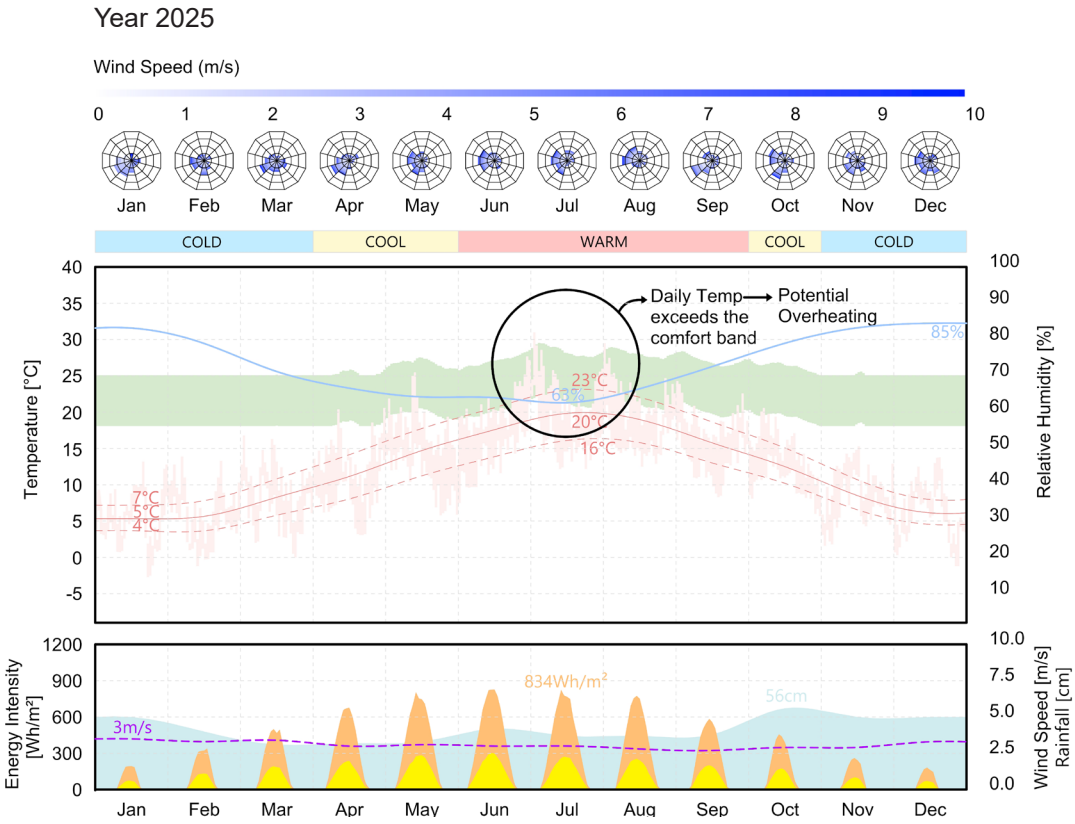


Figure 2.5.2- Climate Analysis- Source- Ladybug Tools

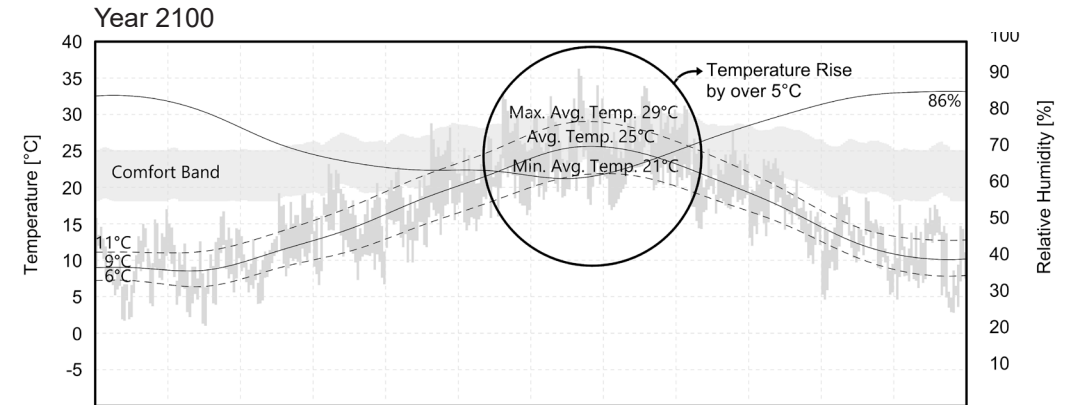
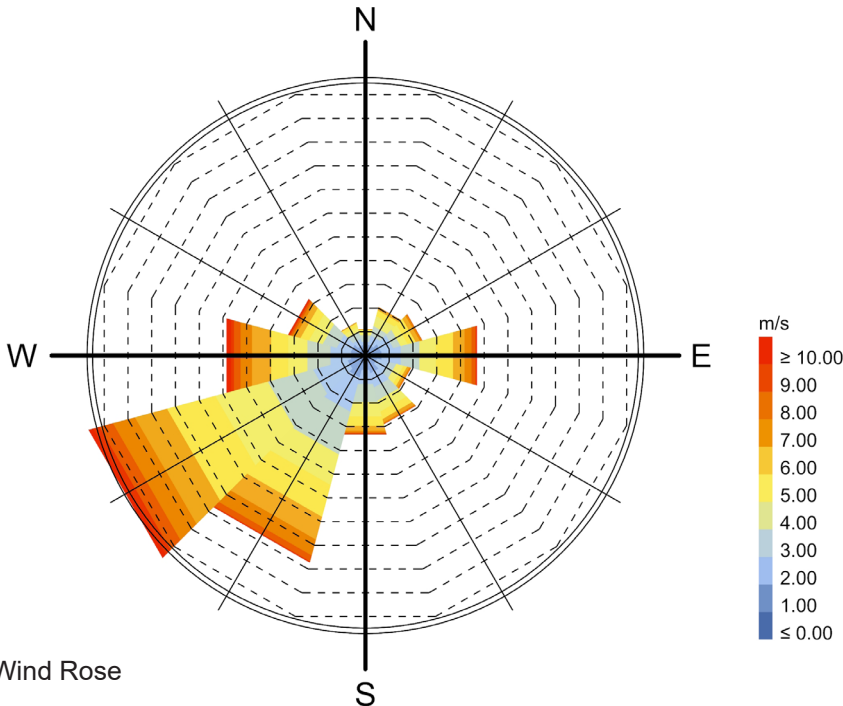


Figure 2.5.3- Future Climate Predictions- Source- Ladybug Tools



Annual Wind Rose
Figure 2.5.4- Windrose Diagram- Source- Ladybug Tools

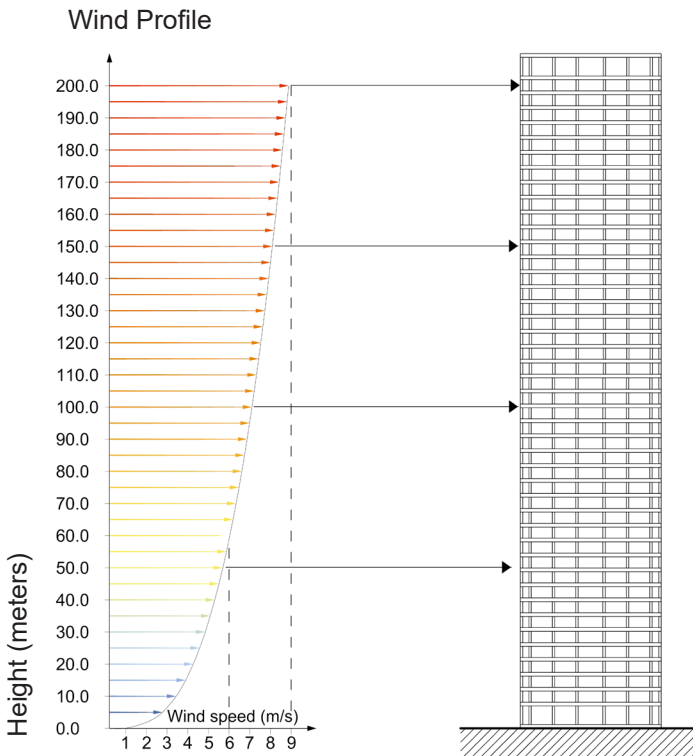


Figure 2.5.5- Wind Profile- Source- Ladybug Tools

3.Pre Design Analytical Design

3.8- Building- Embodied Carbon & U-Values

The total embodied carbon of HSBC over its lifecycle is 840 kg CO2e/m².

HVAC (30% of total embodied carbon)

Life cycle embodied carbon: 205.3 kg CO2e/m²

Operative emissions per year: 80 kg CO2e/m²

Finishes (20% of total embodied carbon)

Suspended plasterboard ceilings: 25 kg CO2e/m²

Raised floors: 150 kg CO2e/m²

Windows (20% of total embodied carbon)

Life cycle embodied carbon: 133 kg CO2e/m²

Superstructure (30% of total embodied carbon):

Life cycle embodied carbon: 158 kg CO2e/m²

Reinforced Concrete Core

Concrete type: RC 32/40 with 25% GGBS

Embodied carbon: 24.4 kg CO2e/m²

Reinforcement: 250 kg/m³

Volume: 18,000 m³

Floor area: 97 m²

Total embodied carbon: 80 kg CO2e/m²

Steel-Frame Slabs

Concrete type: RC 32/40 (100 kg/m³ reinforcement)

Embodied carbon: 89 kg CO2e/m²

Volume: 238,800 m³

Floor area per slab: 3,120 m²

Total embodied carbon: 70 kg CO2e/m²

Steel Columns

Type: UC 254 x 254

Embodied carbon: 22 kg CO2e/m²

Reinforcement: 250 kg/m³

Volume: 3,400 m³

Floor area per column: 0.8 m² (38 columns)

Total embodied carbon: 6.88 kg CO2e/m²

Aluminum Frames

2mm thick hollow aluminum frame

Internal webbing: Yes

Embodied carbon: 94.6 kg CO2e/m²

Volume: 650 m³

Total area: 25,000 m²

Total embodied carbon: 35 kg CO2e/m²

Double Glazed Windows

Two panes per window, 6mm glass

Embodied carbon: 37.9 kg CO2e/m²

Volume: 107,500 m³

Window area: 47,200 m² (5,800 windows)

Total embodied carbon: 98 kg CO2e/m²

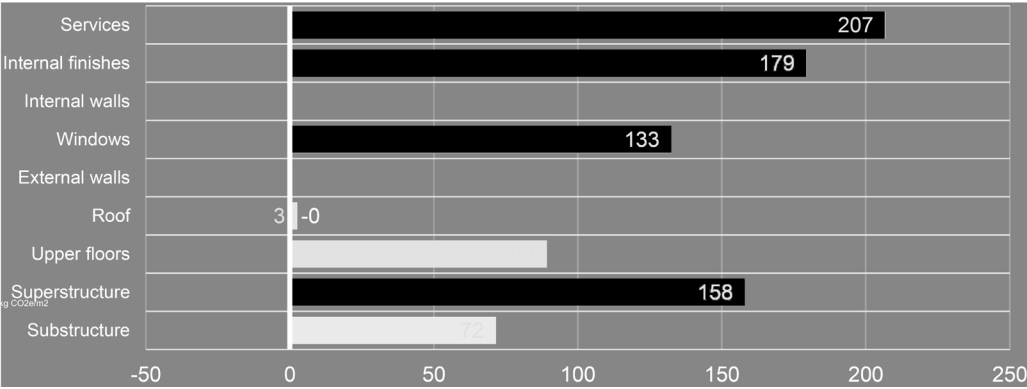


Figure 3.8.1- Existing Building Carbon Analysis Source- FCBS Tool

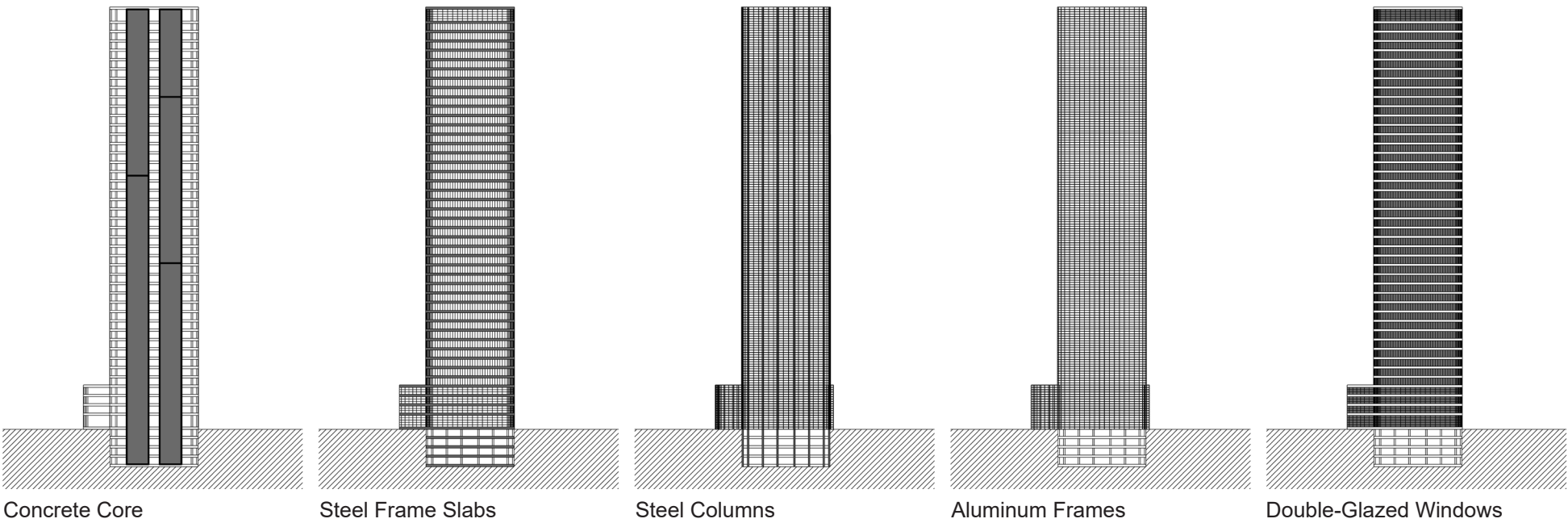


Figure 3.8.2- Building Elements Source- Author

4.Building and Context Analysis

4.1- Performance Targets

To define a design proposal for the HSBC building, the methodology involved analyzing the current conditions of the case study to establish a series of targets and design intentions shaping the proposal.

The diagnosis of the case study's problems and potentials emerged from a series of analyses and simulations: sun path, overshadowing, shadow mask, radiation, daylight, wind studies, Universal Thermal Climate Index, and thermal simulations.

These investigations were key to identifying the main challenges and opportunities of a building of this kind. This chapter examines each parameter individually and in relation to other conditions of the case study and its context.

The final aim is to assess the current performance of the HSBC building and project its improvement. More specifically, the goal is to understand to what extent this tower could perform on a free-running basis — in which seasons, for which programs, and under what conditions. Additionally, if external equipment is needed, the objective is to determine when and why.

4.2- Sun Path

The sun path was analyzed using the center of the case study as a reference point. The HSBC building is surrounded by high-rise towers on the East, South, and West sides, obstructing the sun. In contrast, the lower buildings on the North side do not affect the sun path. Another important consideration is the views: only the North side of the case-study offers uninterrupted views, while the other orientations are compromised by the surrounding towers.

This dense skyline has two main effects: overshadowing on the East, South, and West façades, which is beneficial as it reduces solar exposure; and overshadowing at ground level, which is detrimental as it limits sunlight access. These impacts will be further analyzed.

4.2- Overshadow on Envelope

To assess the shadow situation on the facades and ground floor of HSBC building, three solar studies were conducted: a point-in-time overshadowing analysis on the envelope, a shading mask analysis on the ground floor, and sun access simulations.

The first analysis of the HSBC envelope provided valuable information at three different times of the year (winter solstice, equinoxes, and summer solstice) and three different times of day (morning, noon, and afternoon). This was crucial for assessing the overshadowing percentages on the three surfaces: east, south, and west, and for comparing the conditions of the warm season versus the cold season.

From this, we concluded that the warm season will be most affected by radiation, as it is when the towers cast the least shadow on the facades. In contrast, the cold season has the highest percentage of shadow coverage on the envelope.



Figure 4.1.1- Building and Context Source- Author

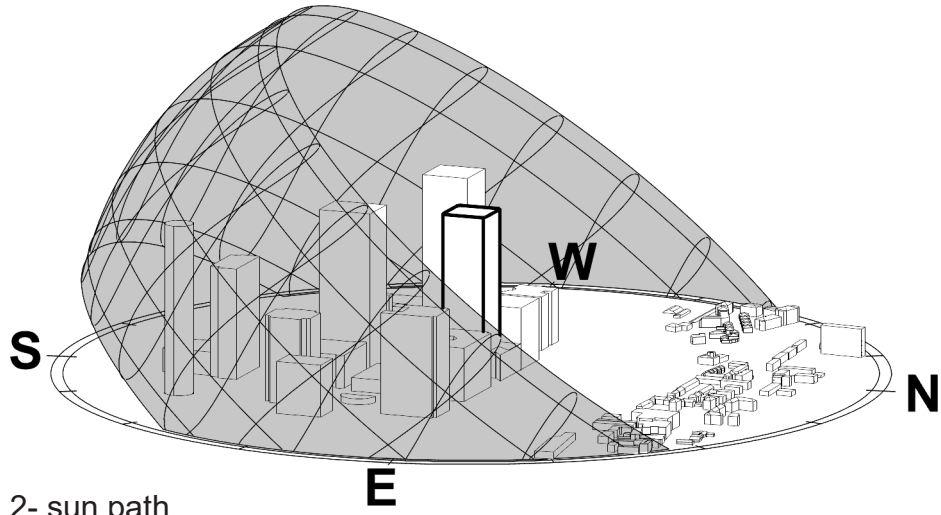


Figure 4.2.1- Sunpath Diagram and Views Source- Lagybug Tools

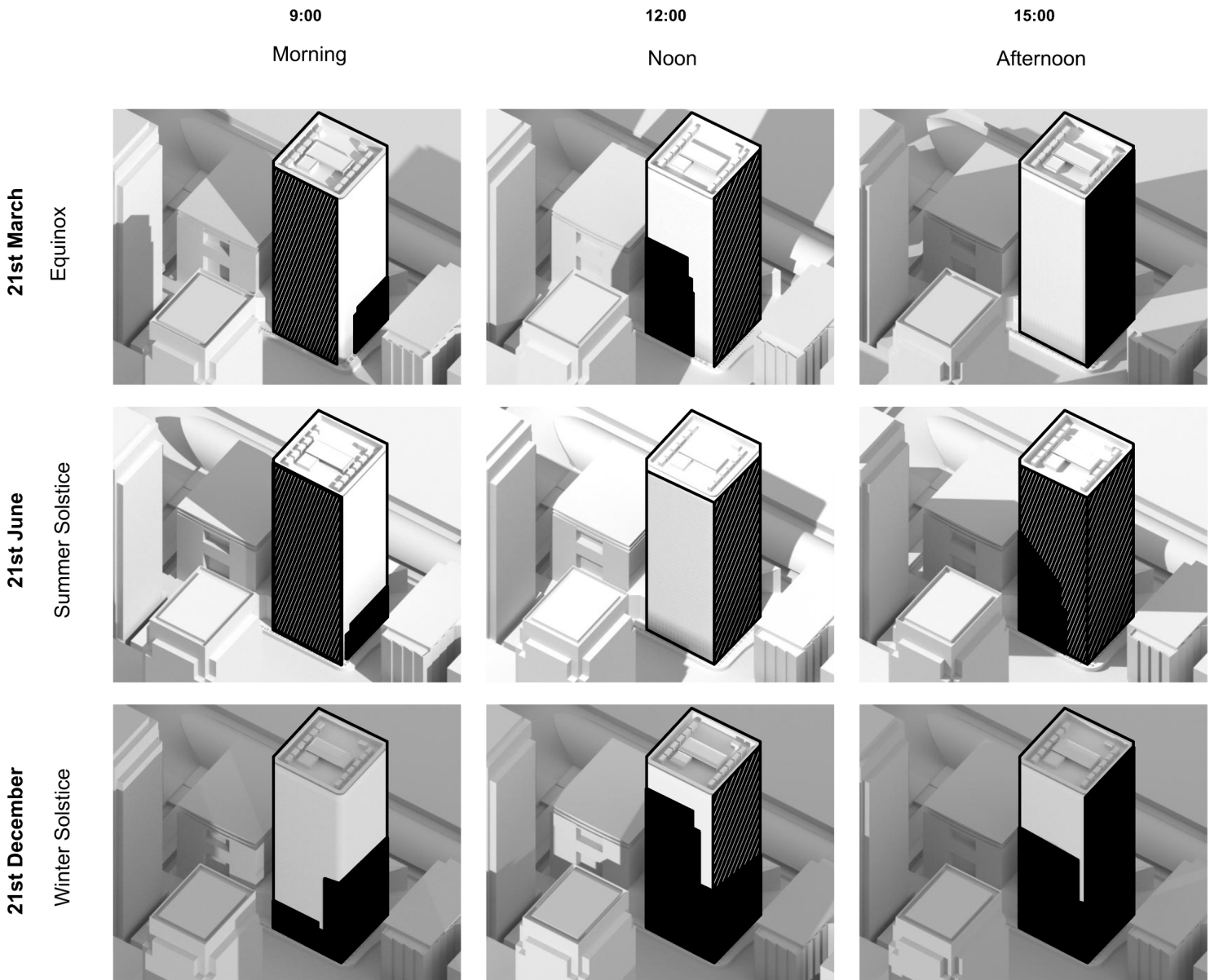


Figure 4.3.1- Overshadow Analysis Source- Rhinoceros

4. Building and Context Analysis

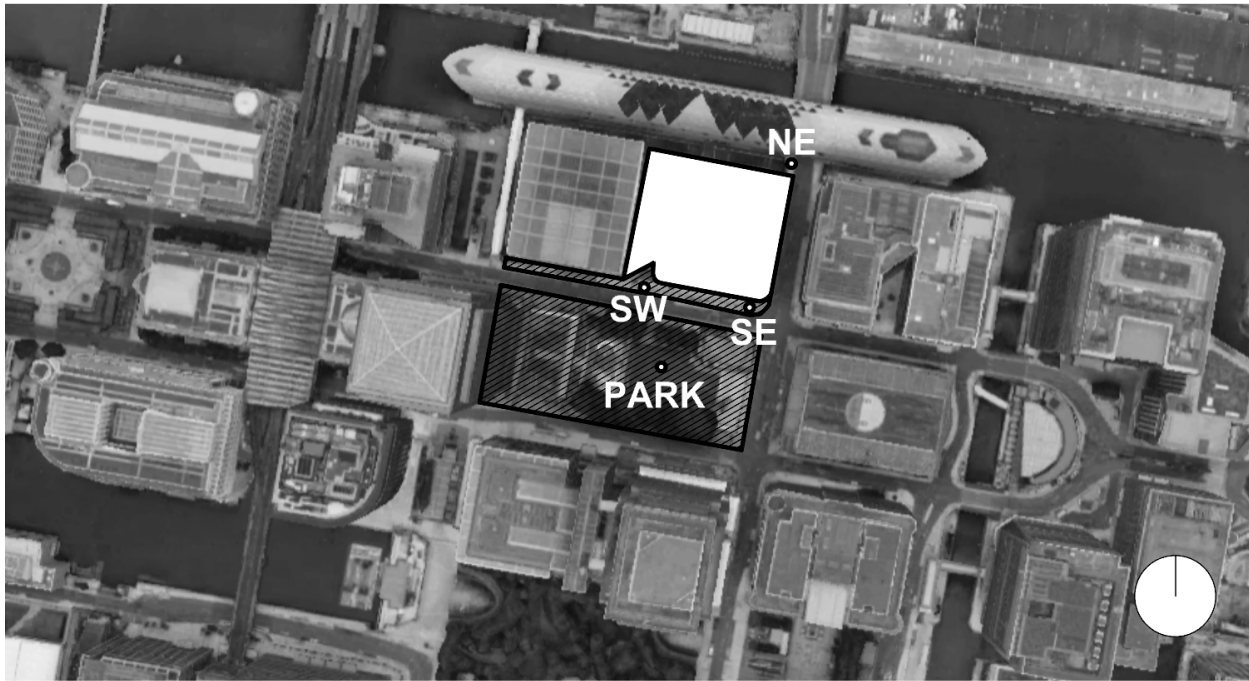


Figure 4.4.1- Shadow Masking Source- Ladybug Tools

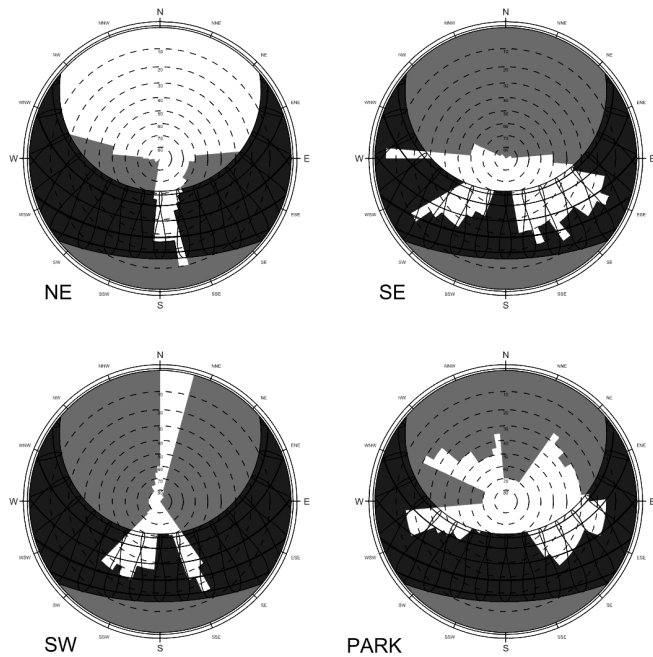


Figure 4.4.2- Shadow Masking Source- Ladybug Tools

4.4- Shading Mask at Ground Floor

In the second sun analysis, the shading mask at ground level was studied in various public areas, such as the park and sidewalks surrounding the case study. It is evident that throughout the year, the sun path is obstructed by the Canary Wharf towers, leading to heavily shaded areas.

4.5- Sunlight Hours

The third and final solar study was the analysis of sunlight hours at the ground floor level, conducted annually.

The sun access in this area is less than 50% of the total daytime hours. This information will inform the proposal by acknowledging that the key strategy will focus on other qualities of the space: instead of assuming a space of constant light, the proposal will suggest a dynamic space characterized by connectivity and movement.

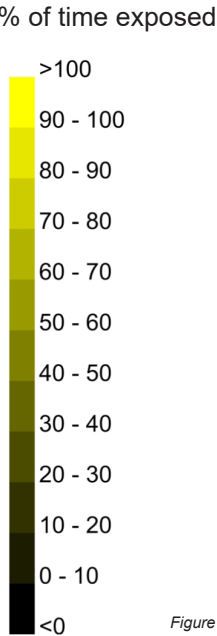
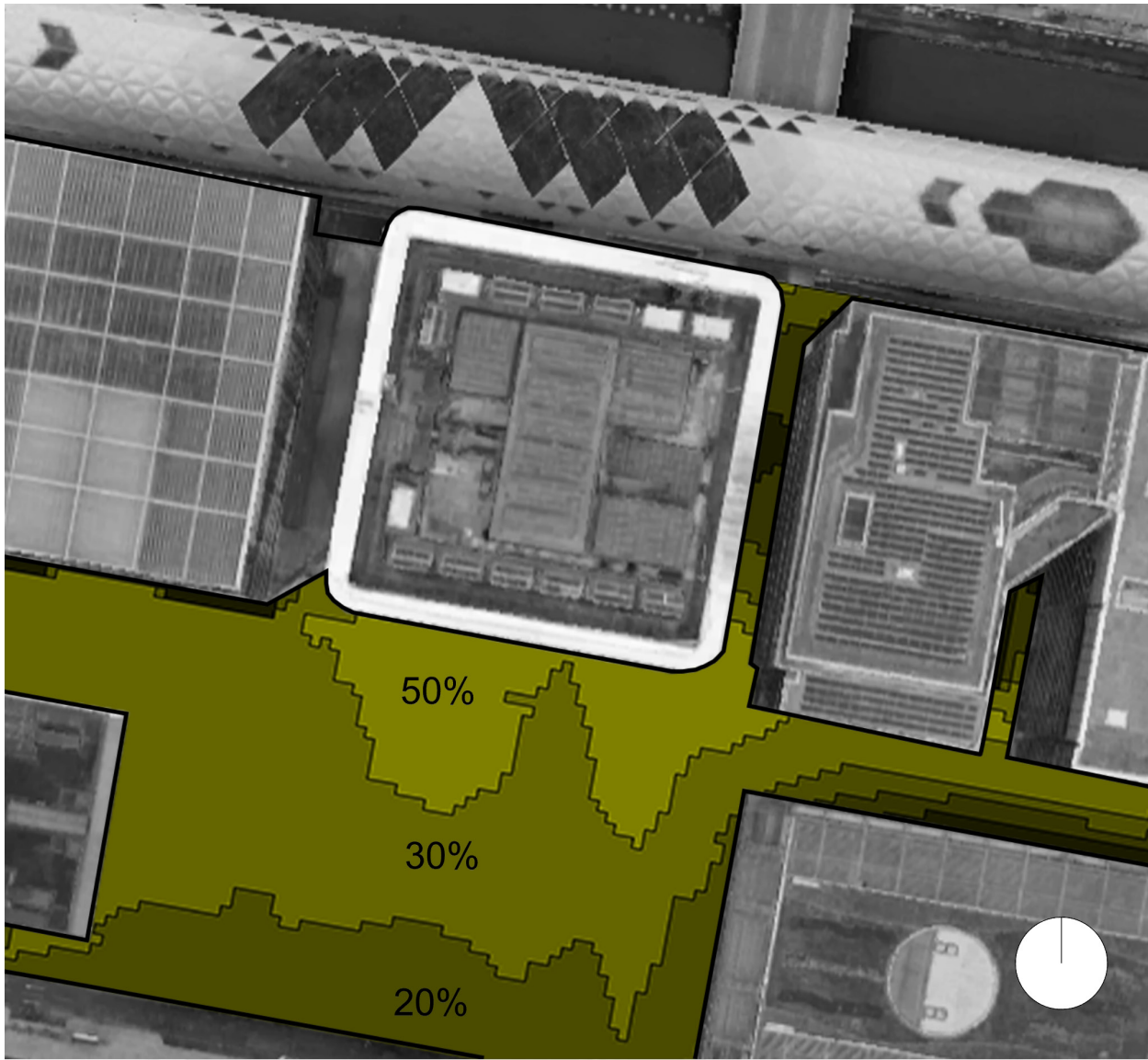


Figure 4.5.1- Sunlight Hours Source- Ladybug Tools

4.Analytical Study
4.6- Daylight

Daylight analysis is included in this Pre-Design Analytical Study chapter. Since building access could not be guaranteed, the analysis is based solely on simulations and assumptions. Given that the building is a fully glazed high-rise tower surrounded by similar structures, daylight availability may vary depending on floor height and orientation. The primary structural elements are the core and columns, with no significant internal partitions.

Two analyses will evaluate daylight within the HSBC building: a Useful Daylight Illuminance study to assess daylight quality for office programs, and a Daylight Autonomy study to determine the need for artificial lighting in residential spaces. Additionally, a Glare study, accompanied by a Reflectance analysis, will provide further insights into addressing glare issues on the office floors.

4.6- Useful Daylight Illuminances (UDI)

In the existing HSBC building, all floor plans—except for the underground levels and ground floor—are offices. However, site studies and surveys have shown that this program distribution is no longer strategic or in high demand. Therefore, this study will analyze daylight conditions for both office and residential spaces.

For commercial (office) programs, a Useful Daylight Illuminance (UDI) analysis was performed to identify areas—by position and size—where adequate natural lighting is achieved without artificial lighting or shading devices. The analysis focused on the lower floors, which are most affected, and revealed that between the 6th and 15th floors, 85% of the available space does not meet the required daylight standards. This is due to insufficient daylight at the back of the offices and excessive daylight near the windows, resulting in glare. Only the north side, where there is no direct sunlight, receives good quality daylight in a small portion of the area (15% of the total available space).

Additionally, a simulation incorporating internal partitions was conducted to explore layout possibilities. The results indicate that subdivisions with a depth of 8 meters and a width of 4 meters do not significantly affect the daylight distribution.

4.6- Daylight Autonomy (DA)

For residential programs, a Daylight Autonomy (DA) analysis was conducted to identify areas—by position and size—where artificial lighting is required. The analysis focused on the mid to upper floors and revealed that 90% of the available space meets the minimum daylight requirements for housing programs. The entire floor plan can be occupied without artificial lighting if no internal partitions are present.

However, when the space is partitioned, the situation changes. In subspaces with a width of 4 meters, some areas begin to lack sufficient daylight. While this subdivision measure is acceptable on its own, issues arise if the space is divided into more rooms—a factor that the proposal will take into account.

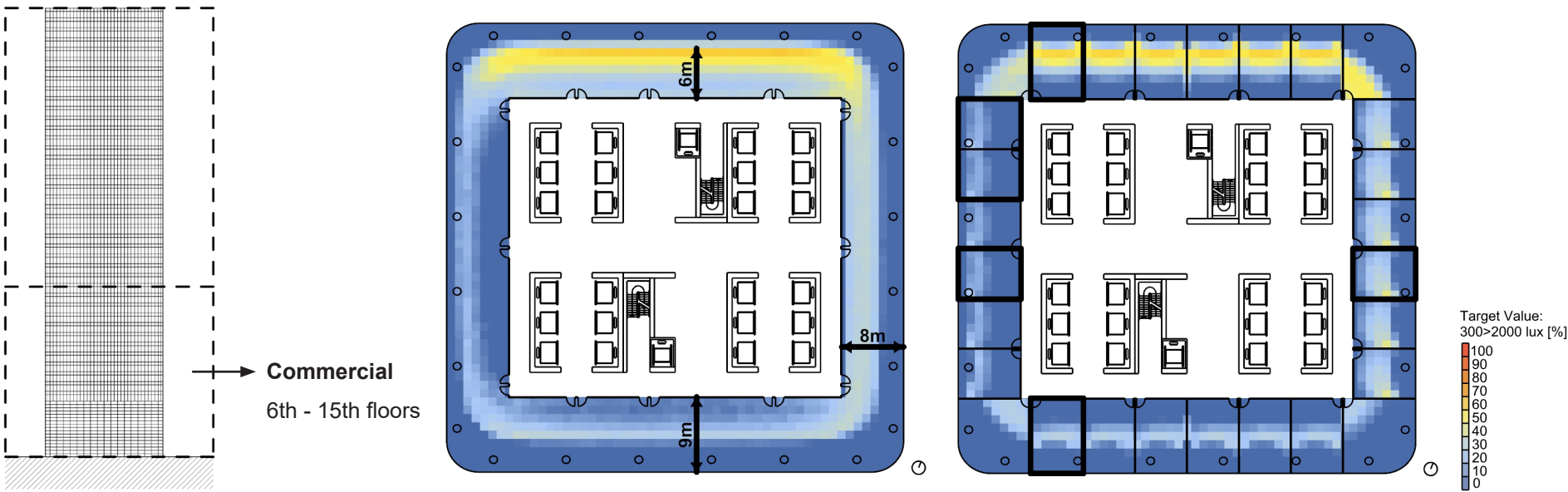


Figure 4.6.1- Daylight Analysis Commercial Source- Honeybee Radiance Tools

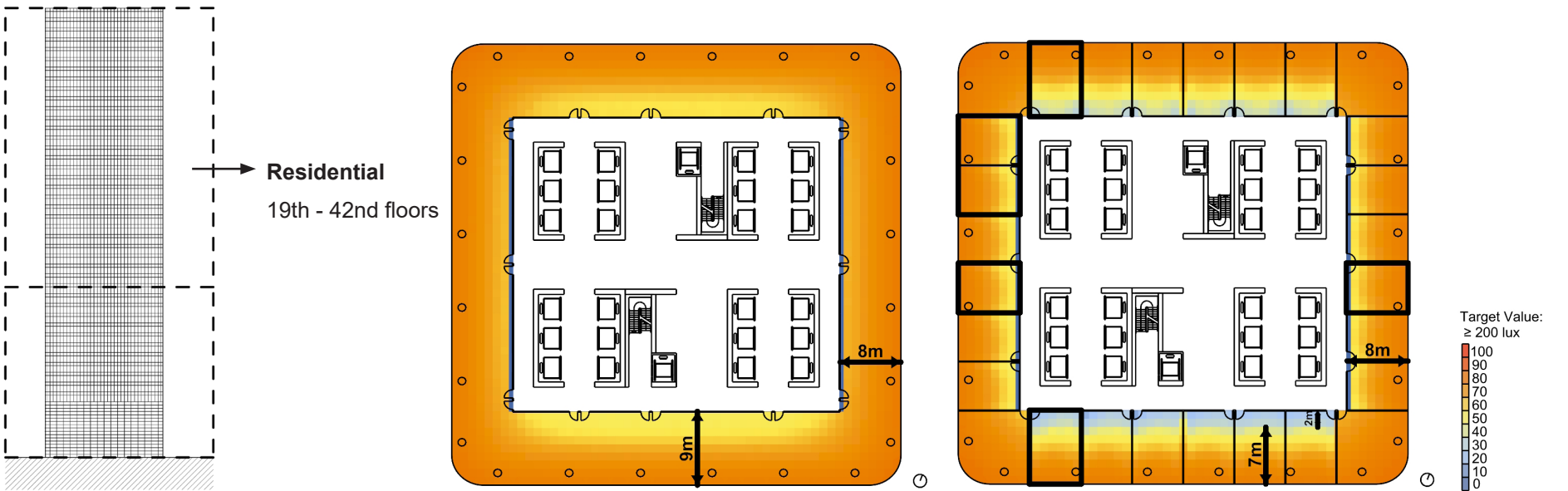


Figure 4.6.2- Daylight Analysis Residential Source- Honeybee Radiance Tools

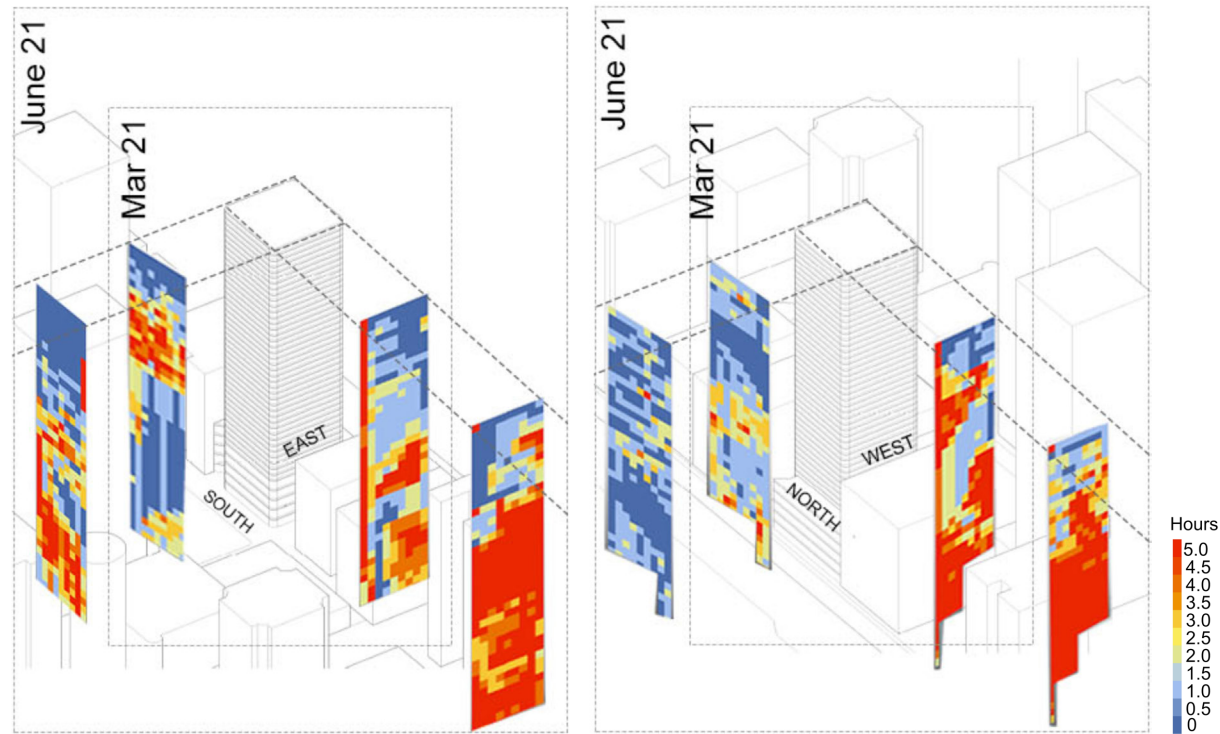


Figure 4.6.3- Reflectance Source- Ladybug Tools

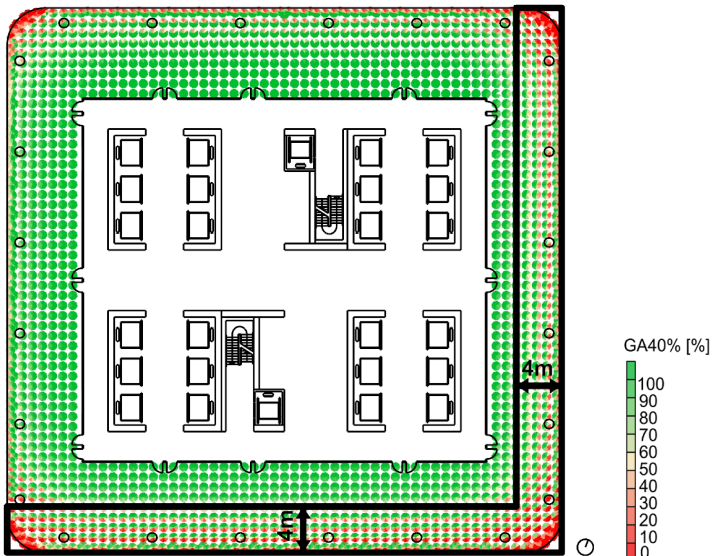


Figure 4.6.4- Glare Autonomy Source- Honeybee Radiance Tools

4.6- Reflectance

Canary Wharf is characterized by glazed towers with reflective facades concentrated in a relatively small floor area compared to their height. Reflectance studies on the HSBC building illustrate the extent of this effect on the case study. Over 60% of the west façade, 70% of the east façade, and 20% of the south façade receive more than 5 hours of reflections daily, contributing to both overheating issues and problematic glare.

4.6- Glare

For the office spaces, an Annual Glare Probability analysis was conducted to identify areas with glare issues. The east and south-facing areas are the most affected, with over 150 m² of the floor plan experiencing glare. Half of the east and south sides spaces have inadequate daylight conditions due to this issue. These results align with the reflectance studies.

This problem will be addressed in the proposal stage by suggesting layout adjustments and offsets to mitigate the glare issue on commercial floors.

4.7- Radiation

To assess the impact of the sun on the HSBC building, two analyses were conducted: a Cumulative Solar Radiation study and a Typical-Day Solar Radiation study. The first study provided an overview of the annual radiation levels. It identified the warm season as problematic, with high radiation levels primarily on the east and south facades. While, during the cold season, the north and west facades receive almost no radiation (close to 0 kWh/m²).

Therefore, the proposal will focus on reducing radiation levels to minimize internal heat gains from solar exposure. Additionally, it will leverage the building's thermal mass to store heat and release it at night.

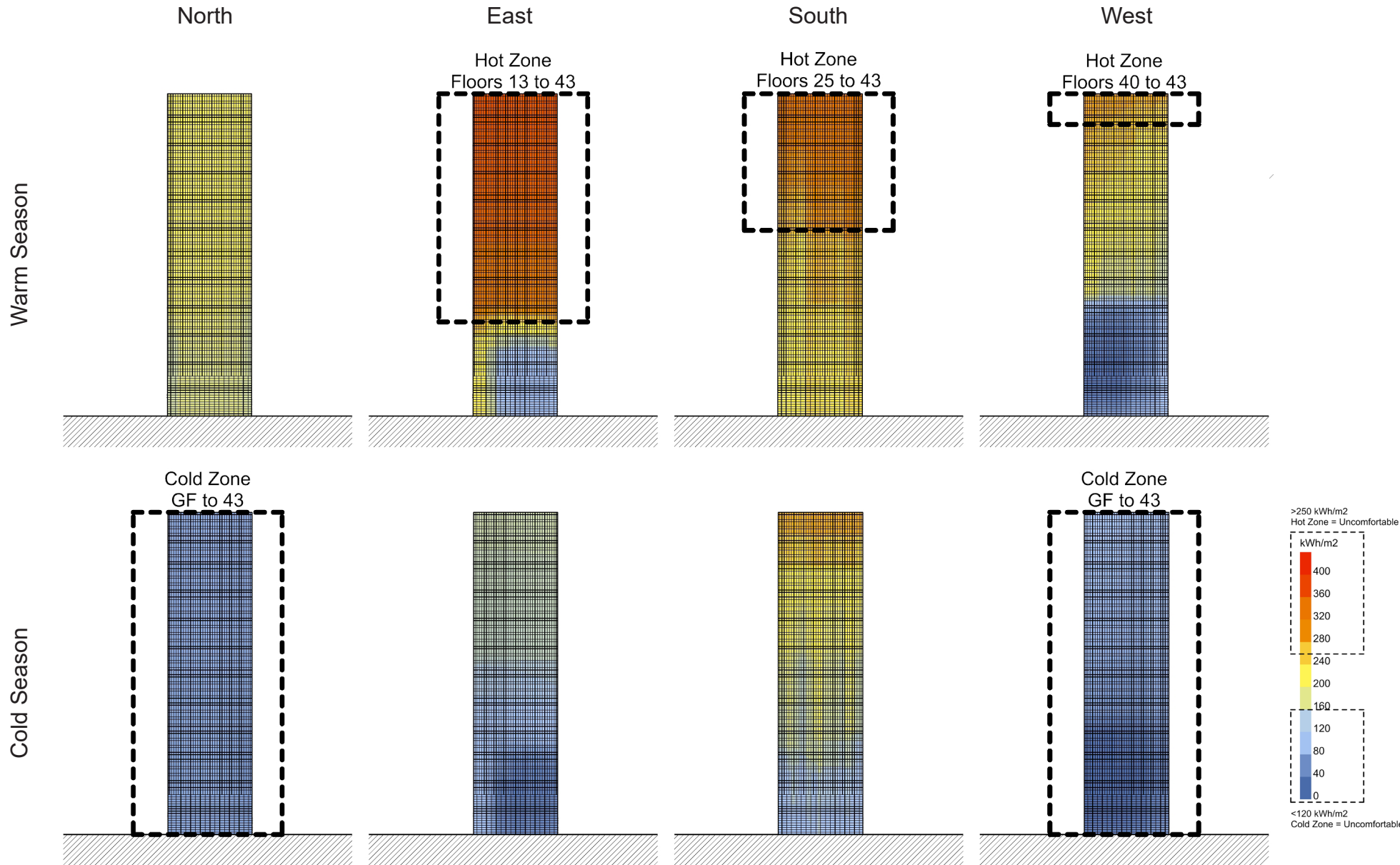


Figure 4.7.1- Solar Radiation on Facade in Summers and Winters Source- Ladybug Tools

4.Analytical Study

4.8- Warm Season Typical-Day Solar Radiation

Since radiation levels during the warm season can cause overheating, a new analysis was conducted to precisely identify the most exposed floor levels. A Typical-Day Solar Radiation analysis for the warm season was performed, using kWh/m² values as input for a MINT analysis to determine the exact radiation levels that lead to overheated spaces.

Four typical office spaces were assessed:
 100 m² office spaces with 12 people and 150 m² glazed area.
 100 m² office spaces with 8 people and 150 m² glazed area.
 80 m² office spaces with 6 people and 100 m² glazed area.
 60 m² office spaces with 4 people and 60 m² glazed area.

For all cases, the following conditions were assumed:
 Ceiling Height: 3 m
 Outdoor Temperature: 20 °C
 Occupants per Floor: Between 1,000 and 1,500
 Equipment: 1 light and 1 computer per person
 Natural Ventilation: No
 HVAC: 8 ACH (ASHRAE 62: 8 L/s per person ~10,000 L/s per office floor)
 Windows: Double-glazed (2.50 W/m²·K)

The analysis concluded that radiation levels over 2.0 kWh/m² can raise interior temperatures beyond the comfort range. Consequently, the south-facing sides of floors 13 and above, as well as the south-facing sides of floors 25 and above, are particularly compromised. These areas must be redesigned in the proposal stage to reduce radiation levels on the windows.

4.9- Thermal Analysis

Once the impact of radiation on the interior temperature of the HSBC building was thoroughly understood, a Thermal Analysis was conducted to identify other factors contributing to the overheating problem. Unsurprisingly, electrical equipment and the lighting system were identified as major contributors to internal heat gains.

For office spaces, reducing the use of electrical equipment is unlikely. However, artificial lighting usage could be significantly reduced if better daylight quality were available in the interior spaces. This is something the proposal will later demonstrate.

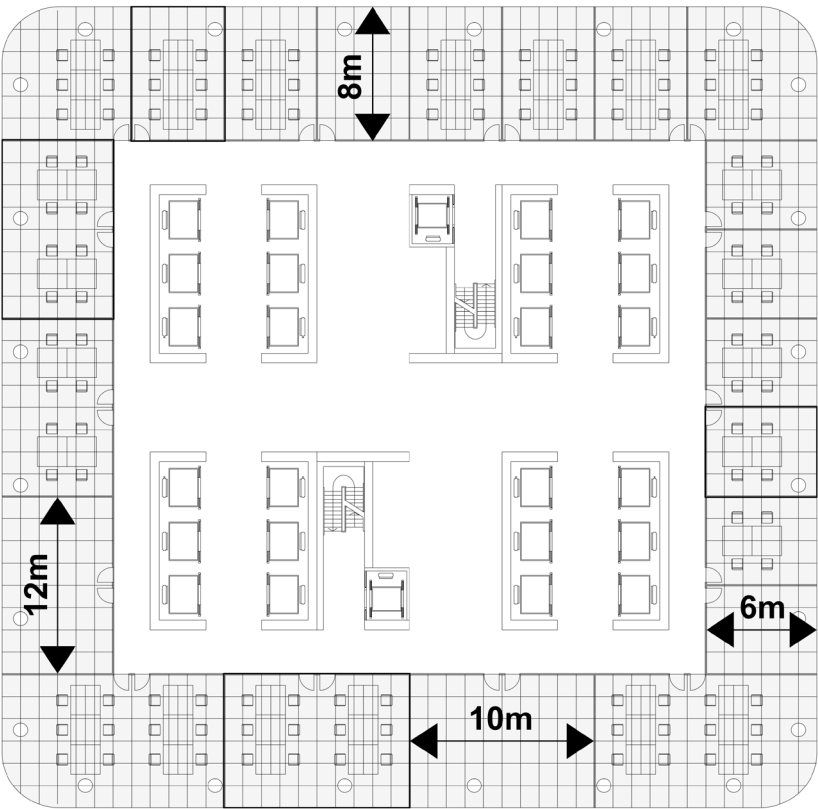


Figure 4.8.1- Base Case Options for office Source- Author

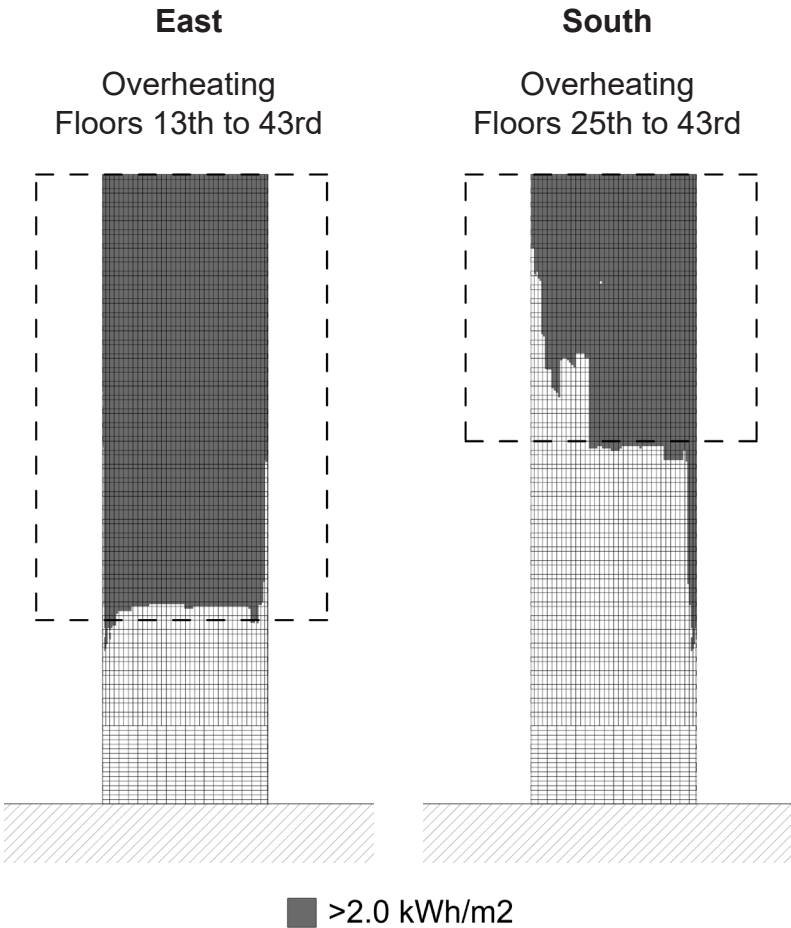


Figure 4.8.2- Overheating Potential Source- Honeybee Radiance Tools

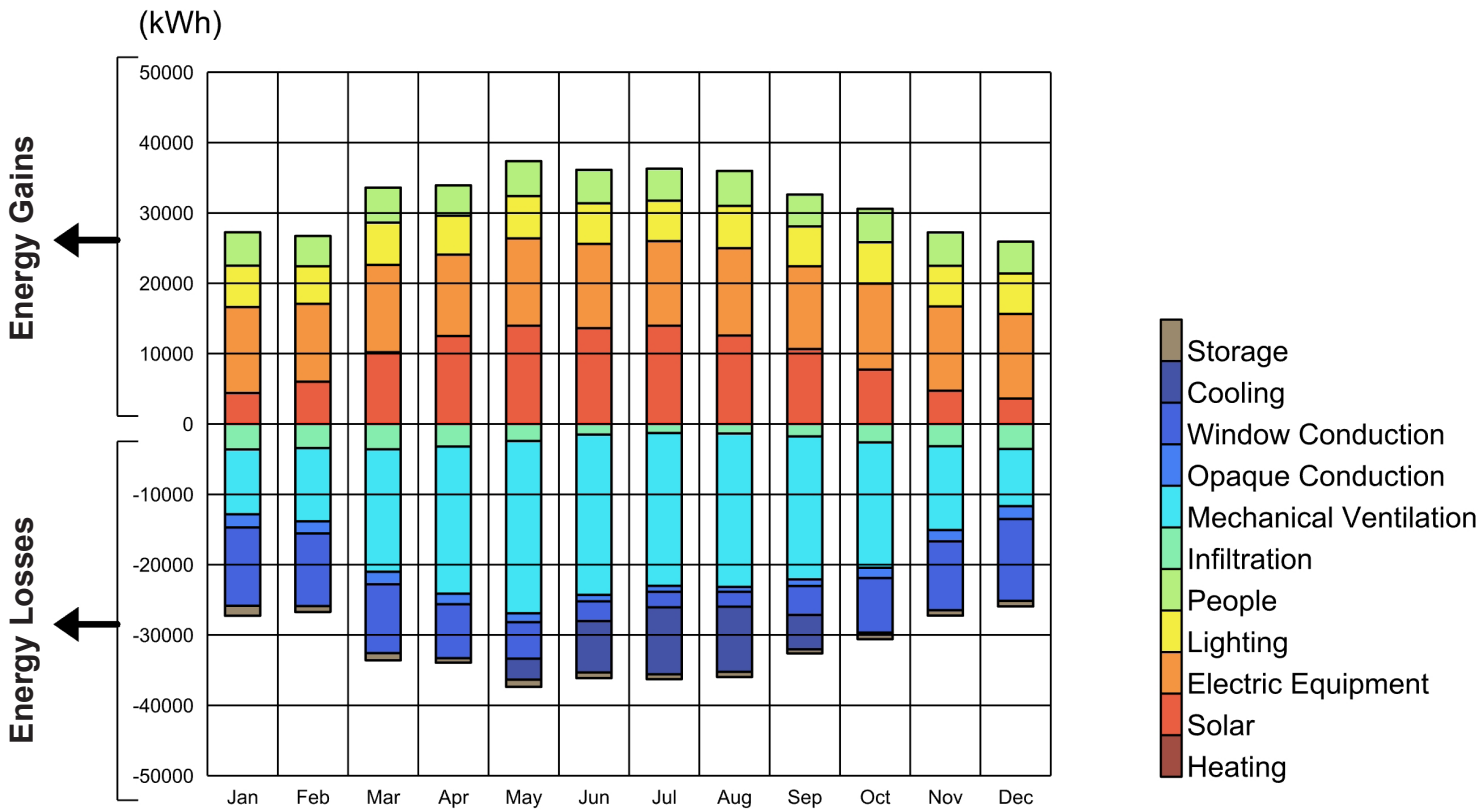


Figure 4.9.1- Thermal Analysis Base Case Source- EnergyPlus

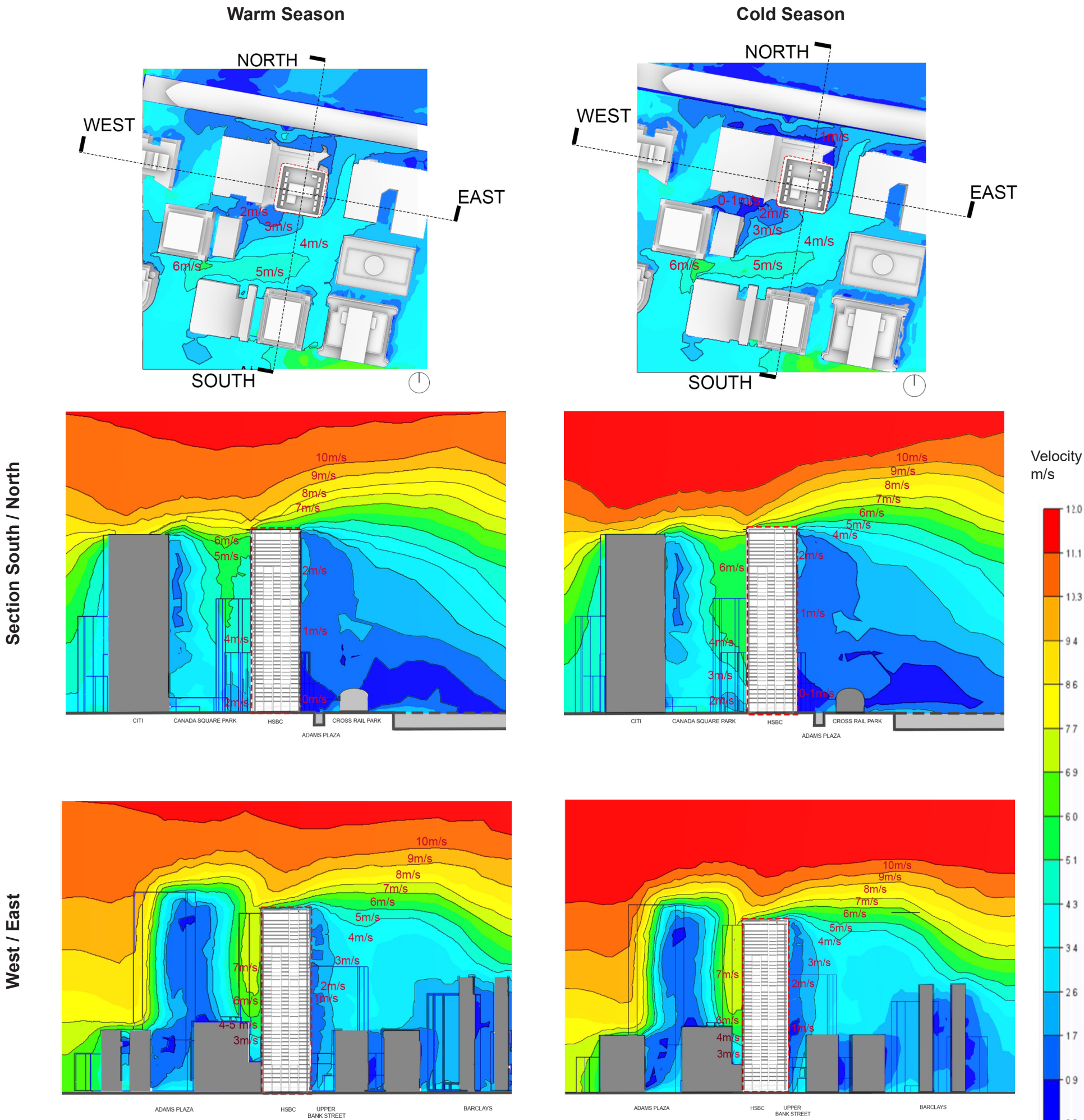


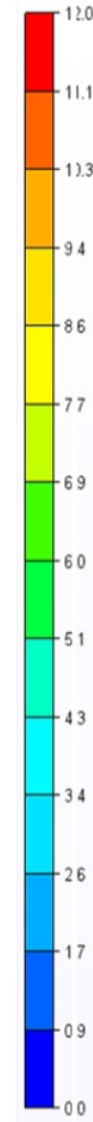
Figure 4.10.1- Wind Studies Source- Autodesk CFD

Due to the height of this 43-storey high-rise tower, it is extremely exposed to wind influx, especially at higher levels. Annually, winds predominantly come from the southwest, making these facades the most exposed to wind speeds exceeding 5 m/s, while the north and east facades experience negative pressure, with average wind speeds of around 2 m/s.

At the ground level, most areas experience medium wind speeds of approximately 4 m/s, but some areas, particularly the park next to the case study, are subjected to strong winds over 5 m/s.

The proposal will pay special attention to the positive and negative pressure differences across the HSBC building's facades. As providing natural ventilation is a key objective to reduce overheating and improve user comfort, it will be necessary to reduce wind speeds on the highly exposed south and west sides to a maximum of 4 m/s.

Velocity
m/s



> 4 m/s	< 5%	Sitting
> 6 m/s	< 5%	Standing
> 8 m/s	< 5%	Strolling
> 10 m/s	< 5%	Business Walking
> 10 m/s	> 5%	Uncomfortable
> 15 m/s	> 0.023%	Unsafe frail
> 20 m/s	> 0.023%	Unsafe all

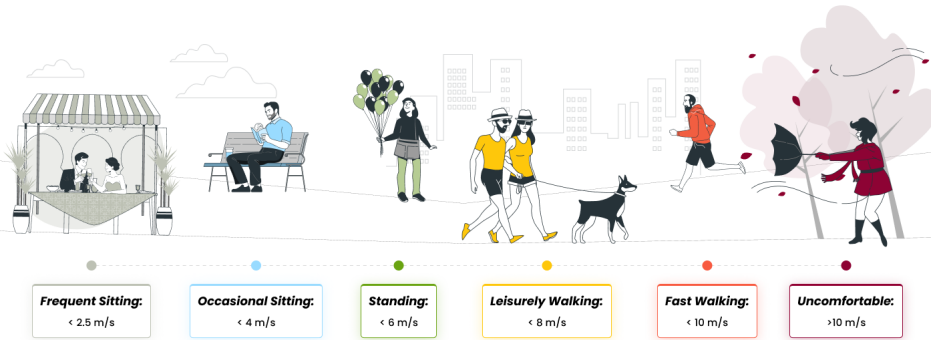


Figure 4.10.2- Wind Comfort Standards Source- Lawson Criteria

5.2. Cluster 1- Offices - Co-working Spaces

Office : Useful Daylight Illuminance

The Useful Daylight Illuminance (UDI) metric was used to evaluate the usable and non-usable areas of the office floor based on daylight availability. Initially, UDI was run on the existing deep floor plan of the HSBC layout, revealing that only 10% of the floor space fell within the usable daylight range. The remaining 90% of the floor plate was either unoccupied or unusable due to excessive glare. This analysis played a key role in strategically placing functions and determining which façades would perform best in terms of daylight access.

A target illuminance range of 300–2000 lux was set, aligning with standard office lighting requirements. To assess the effectiveness of the proposed design in improving daylight distribution, UDI simulations were conducted on both the existing and proposed layouts, as shown in Fig.5.2.10 and Fig. 5.2.11. The results indicated a 65% improvement in daylight performance and space usability compared to the existing layout.

The few areas where glare remained were intentionally carved out as outdoor breakout spaces, ensuring they serve as functional and comfortable extensions of the indoor environment.

Overhangs and punctures were introduced at alternate levels on the south side after analyzing the sunpath, concluding that the slabs act as self-shading elements for the floors below.

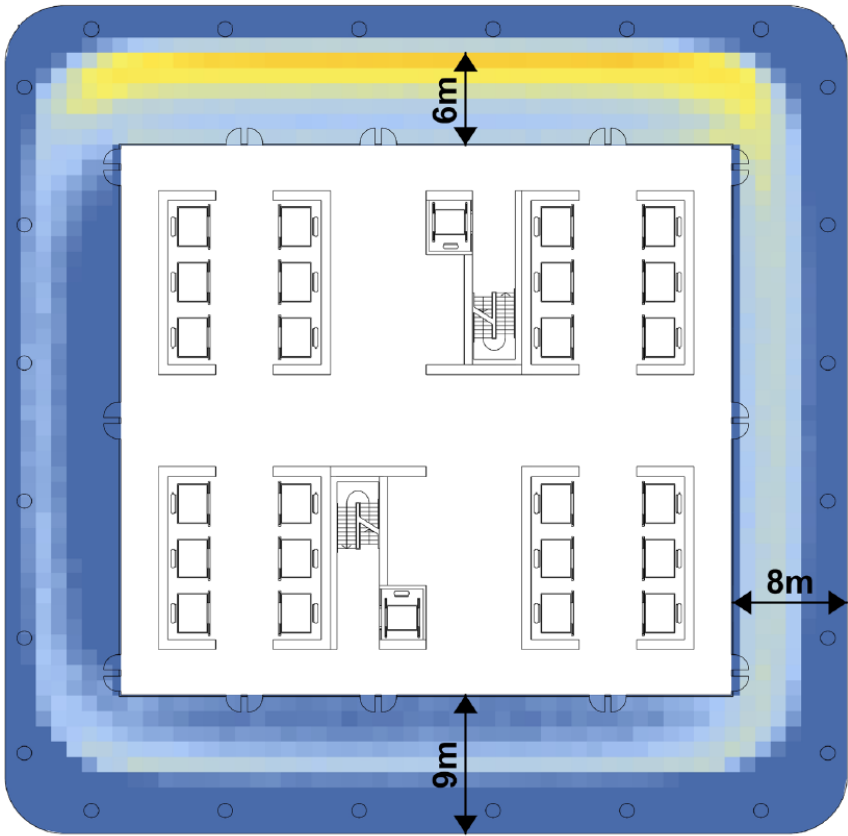


Fig 5.2.10- Existing UDI of a typical office floor Source- Honeybee Radiance Tools

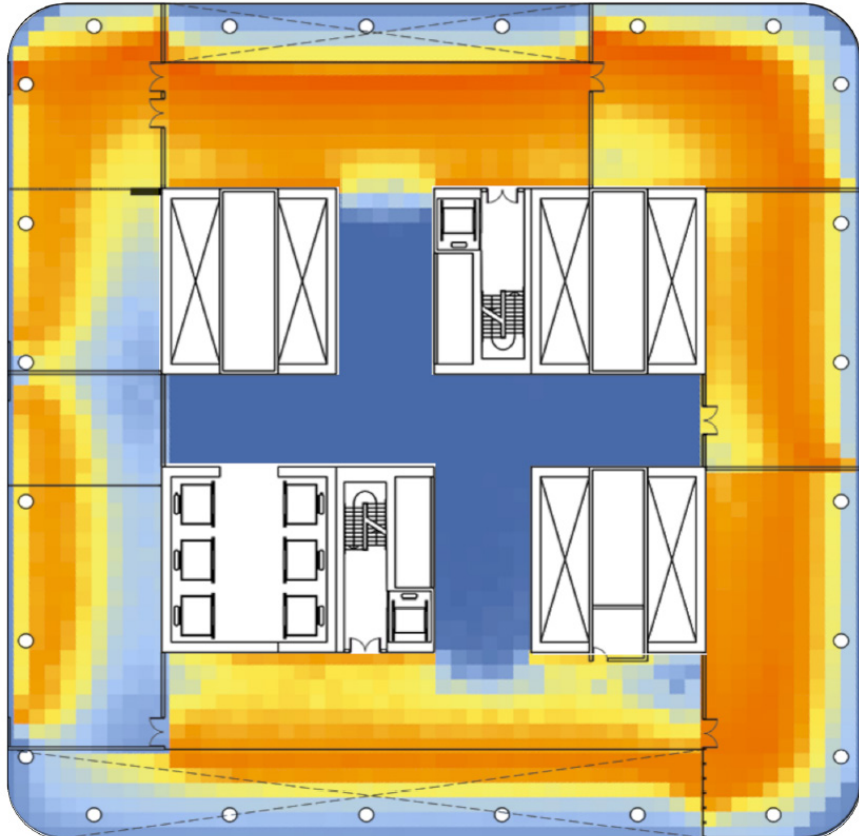


Fig 5.2.11- UDI for the Proposal (9th floor) Source - Honeybee Radiance Tools

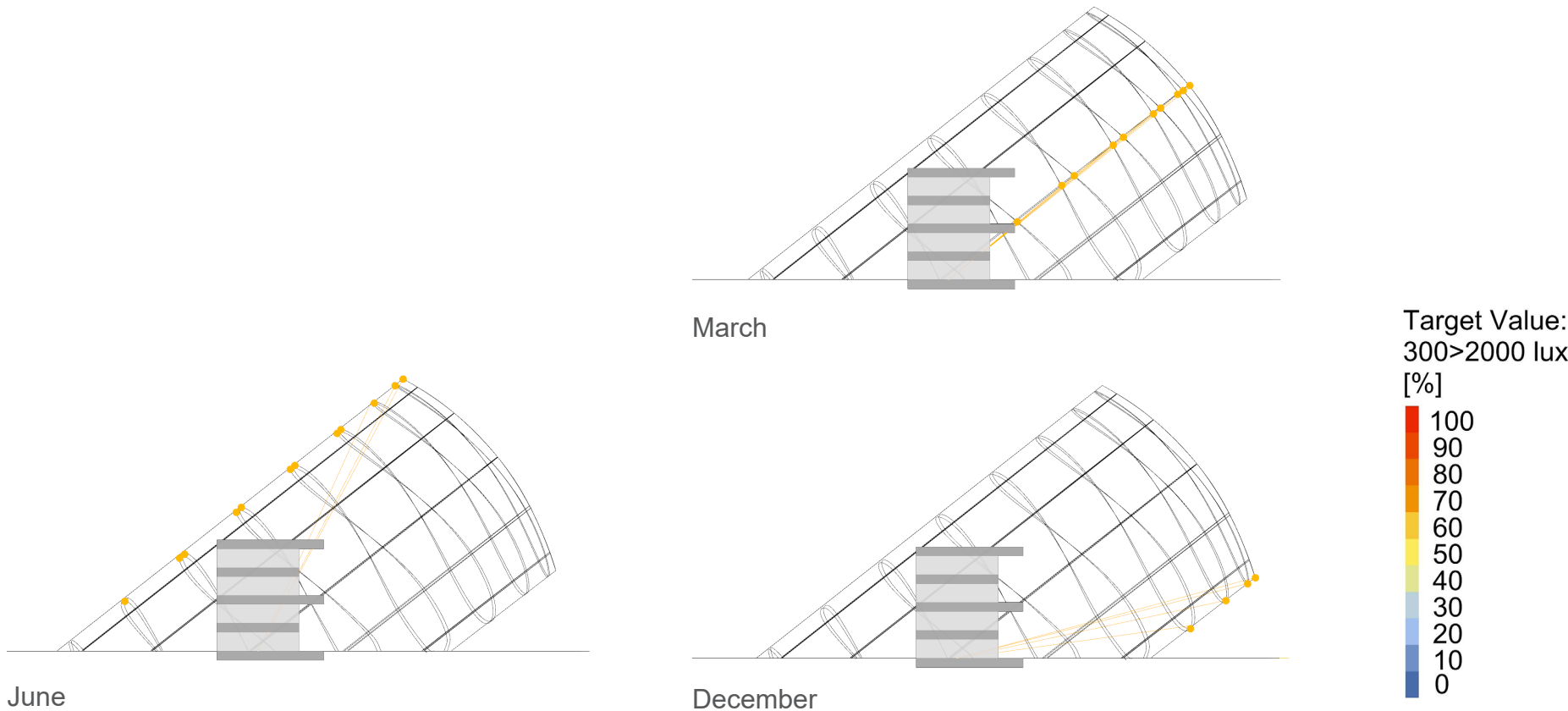


Fig 5.2.12- Sunpath diagrams for the south facade Source- Ladybug

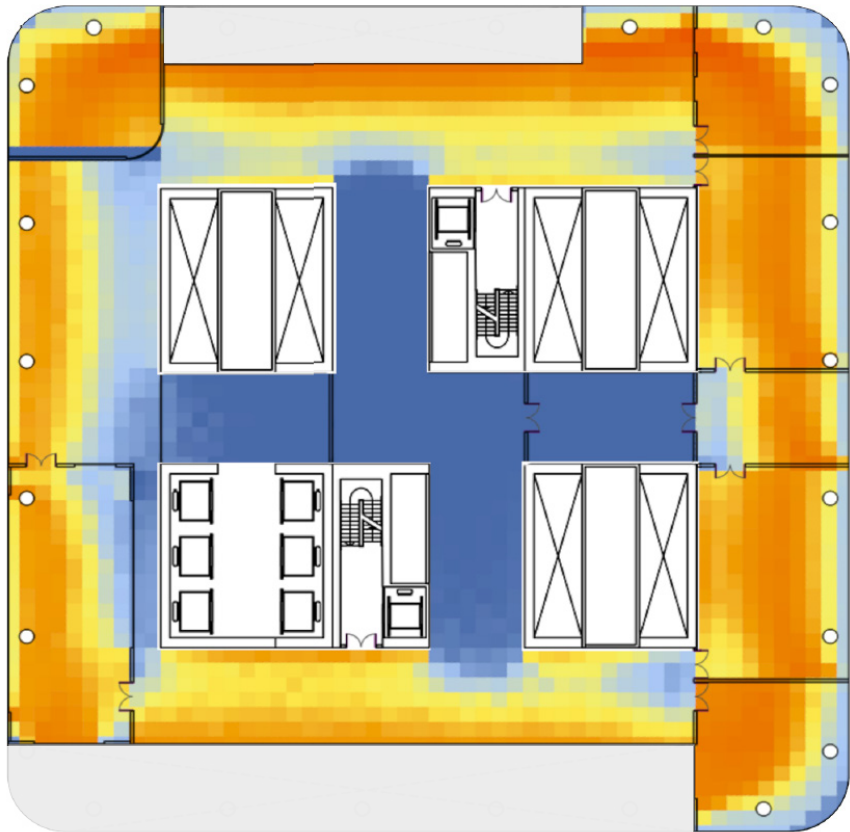


Fig 5.2.13 : UDI for the Proposal (10-14th floor) Source - Honeybee Radiance

5.2. Cluster 1- Offices - Co-working Spaces

Office : Thermal Performance - Base Case Parameters

Following the daylight analysis, a typical bookable room located in the southwest corner was selected to evaluate indoor thermal comfort over a typical summer working week in June.

To further optimize the thermal performance of the building, a base case thermal simulation was conducted as a reference for subsequent simulations. It is important to note that this base case was modeled without any passive strategies to establish a benchmark. The input parameters considered for these simulations are outlined in Table 5.2.14.

Using EnergyPlus, the thermal study was run to assess the annual heating and cooling loads required for the base case. The results indicated:

Heating	load:	12.1	kWh/m ² /year
Cooling	load:	3	kWh/m ² /year

The primary objective is to transition towards a free-running building, significantly reducing or eliminating energy loads. To achieve this, various materials and design parameters were tested and optimized through simulations to meet the targeted thermal performance.

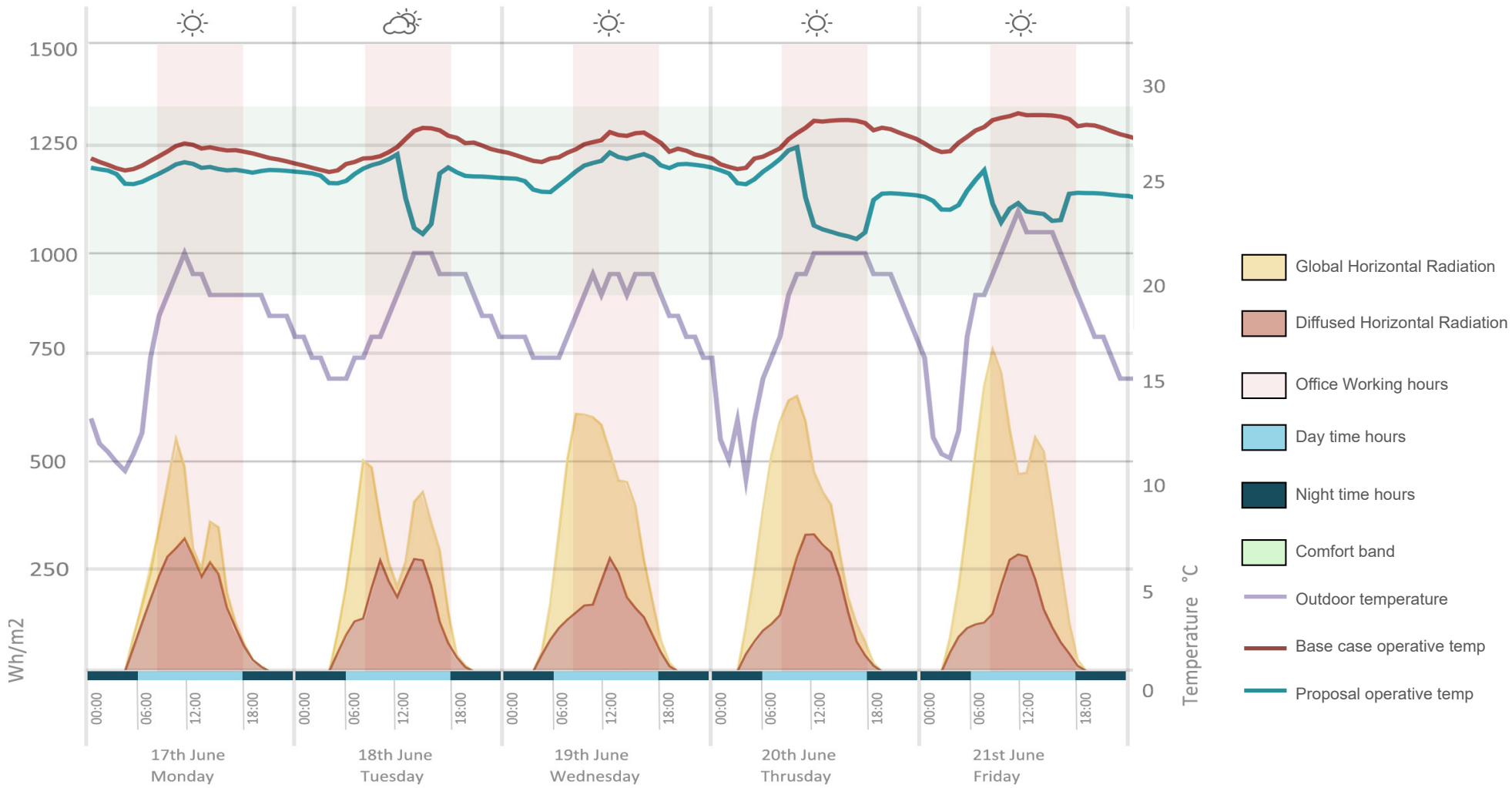


Fig 5.2.14- Operative indoor temperatures during a typical summer working week Source- EnergyPlus

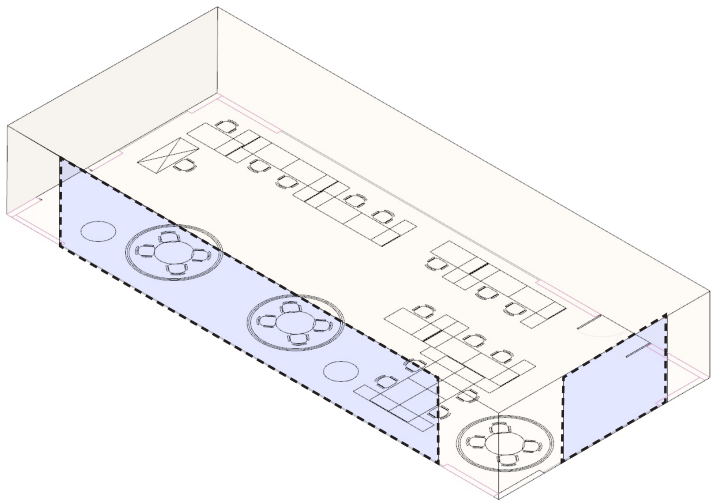


Fig 5.2.16- Typical Bookable room (South-West)

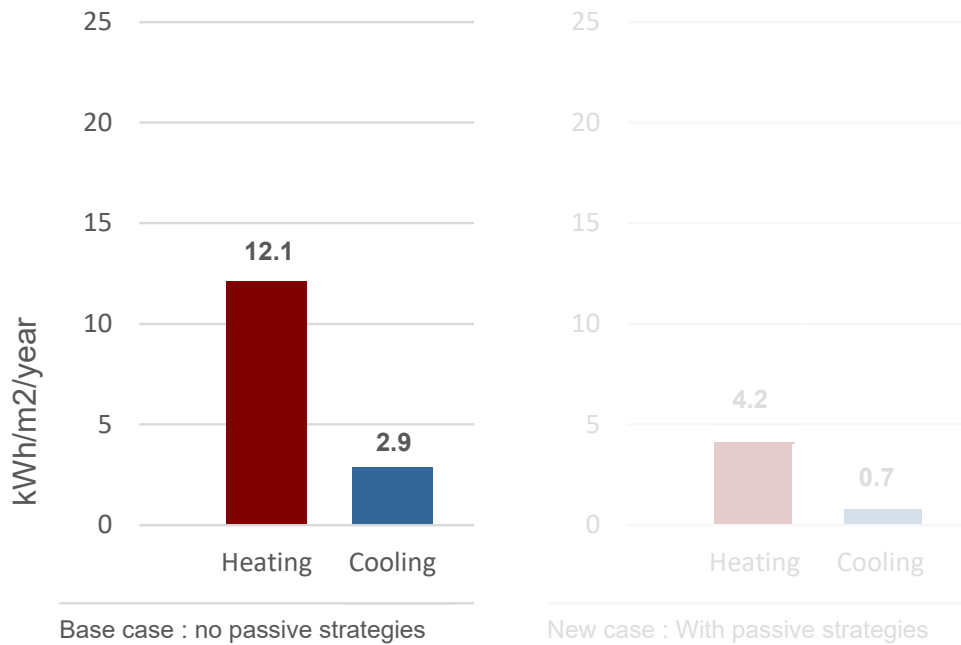


Fig 5.2.15- Comparison of heating & cooling loads of existing and proposed layout Source- EnergyPlus

Input parameters			
Sr. No	Inputs	Value	Units
1	Area	250	sq.m
2	WWR for S	60	%
3	WWR for W	80	%
4	U value Wall	0.3	W/m2K
5	U value Window	1.69	W/m2K
6	Occupants	25	people
7	Lighting Load	6	W/m2K
8	Equipment	6	W/m2K

Table 5.2.17- Input parameters for Thermal Simulations

5.3. Cluster 2- Serviced Apartments

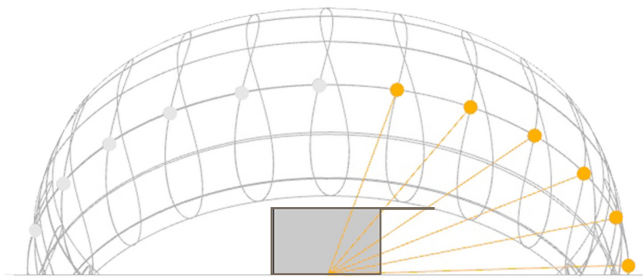
Shading Devices

To improve indoor comfort and energy performance, the building employs a range of strategic shading solutions, particularly on the west and south sides, where sunlight exposure is most intense during the day.

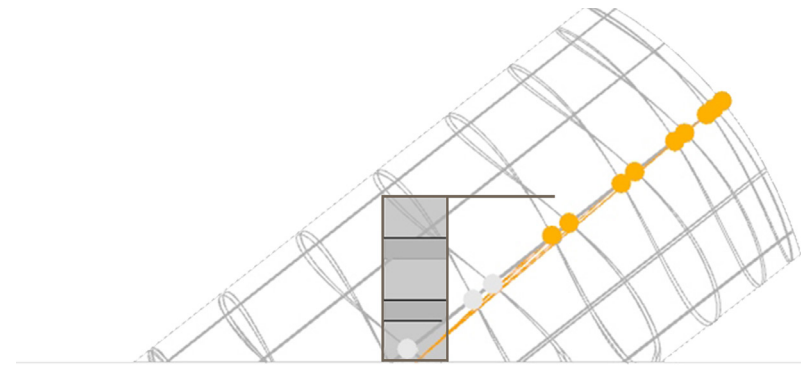
On the west side, horizontal overhangs are incorporated to reduce direct sunlight penetration in the afternoon, preventing overheating while maintaining desirable natural light levels. This strategy is essential for managing the intense solar gain that typically occurs during the late afternoon, which can lead to uncomfortable indoor temperatures. Additionally, vertical shading elements are employed on both the west and south facades to further control solar access and enhance thermal comfort by reducing direct heat absorption.

On the south side, overhangs also play a critical role in regulating solar exposure, allowing for natural light to filter into the building while mitigating excessive heat gain, which can affect energy efficiency. Vertical shading on this side ensures that solar access is balanced, contributing to a comfortable indoor environment year-round.

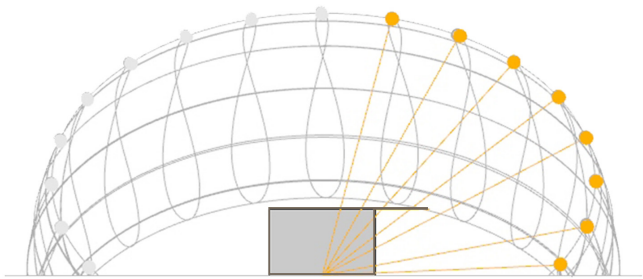
Furthermore, a double facade system is utilized throughout the building to provide thermal insulation and acoustic comfort, further enhancing energy efficiency and occupant well-being. Blinds are integrated as an additional measure for occupants to control solar access based on their preferences, giving them flexibility to adjust the interior environment as needed. These shading solutions are carefully designed to balance daylight, heat control, and comfort, contributing to the overall energy performance and livability of the building.



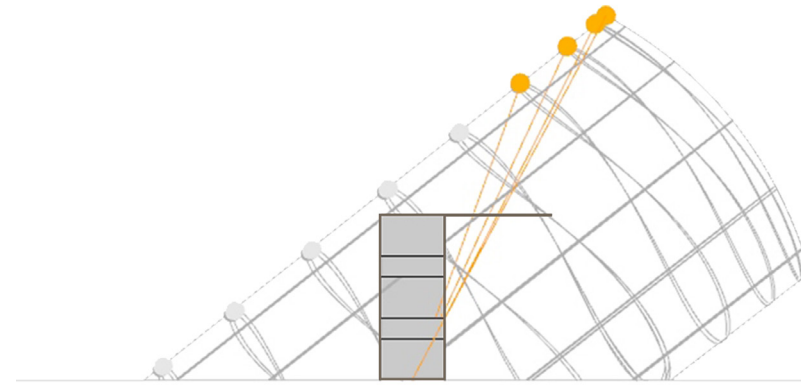
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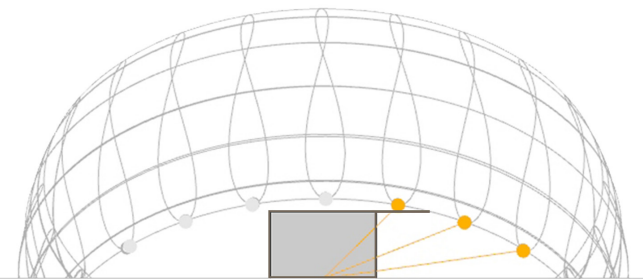
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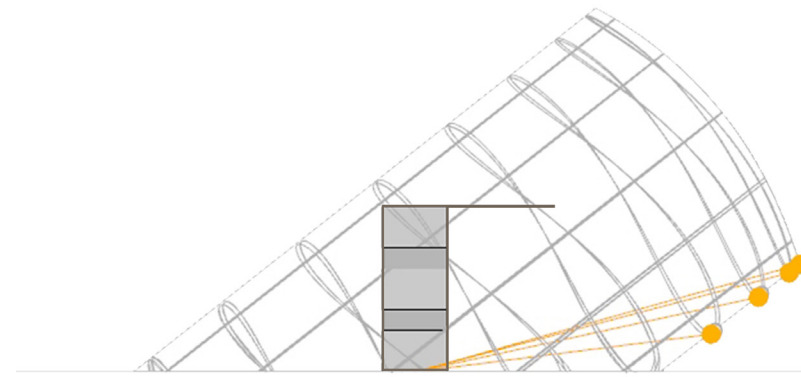
June



June



December



December

Fig 5.3.11- Shading West Source- Ladybug Tools

Fig 5.3.11- Shading South Source- Ladybug Tools

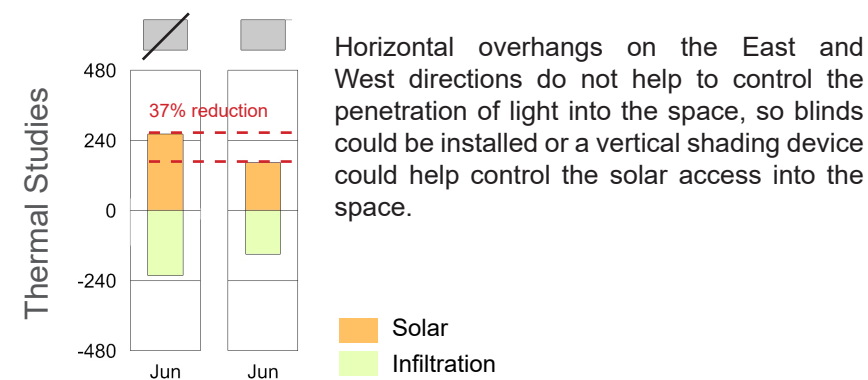


Fig 5.3.11- Shading Studies Source- EnergyPlus

5.3.Cluster 2- Serviced Apartments

Daylight Study

The daylight study explores how different floor layouts and glazing sizes impact daylight distribution within the apartment spaces. The analysis shows that the 19th floor achieves a good spread of daylight, with high daylight autonomy (DA) values along the perimeter and improved light penetration into the core areas, which were previously quite dark. This indicates that the revised layout allows for better light distribution, even in deeper interior zones. In contrast, the typical floor plans on the higher floors (20–26) demonstrate a more controlled daylight spread due to increased shading from adjacent structures and internal elements. Despite this, all units receive sufficient daylight according to their orientation, making them more functional and comfortable.

To refine the design further, different glazing sizes were tested using soft computations to evaluate their impact on daylight levels and thermal comfort. Larger glazing sizes, like the 4.75 m option, allowed more daylight penetration but also increased the risk of glare and overheating. On the other hand, the smallest option (2.40 m) resulted in insufficient daylight, especially in the deeper parts of the rooms. Option 3 (3.00 m) offered a balanced daylight spread but left some uneven lighting in the core areas. Ultimately, Option 2 (3.00 m glazing) emerged as the best solution, striking the right balance between daylight penetration and thermal comfort while ensuring that all functional areas receive adequate natural light. This makes Option 2 the ideal choice for optimizing both daylight performance and indoor comfort.

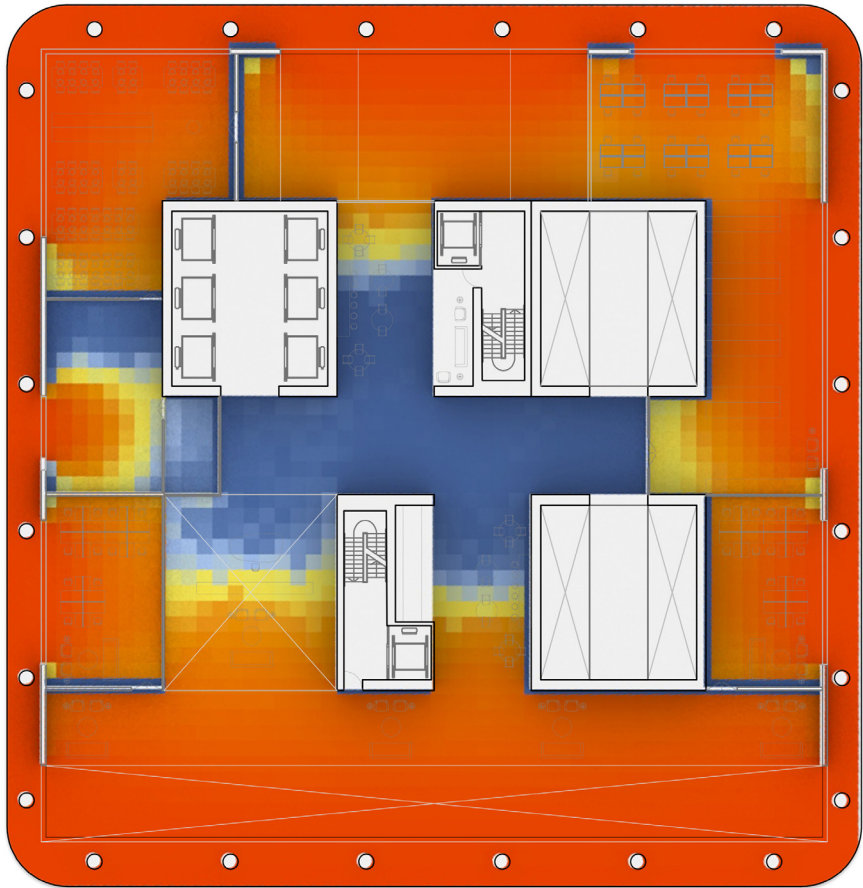


Fig 5.3.11- Daylight Autonomy- Floor 19 Source- Honeybee Radiance

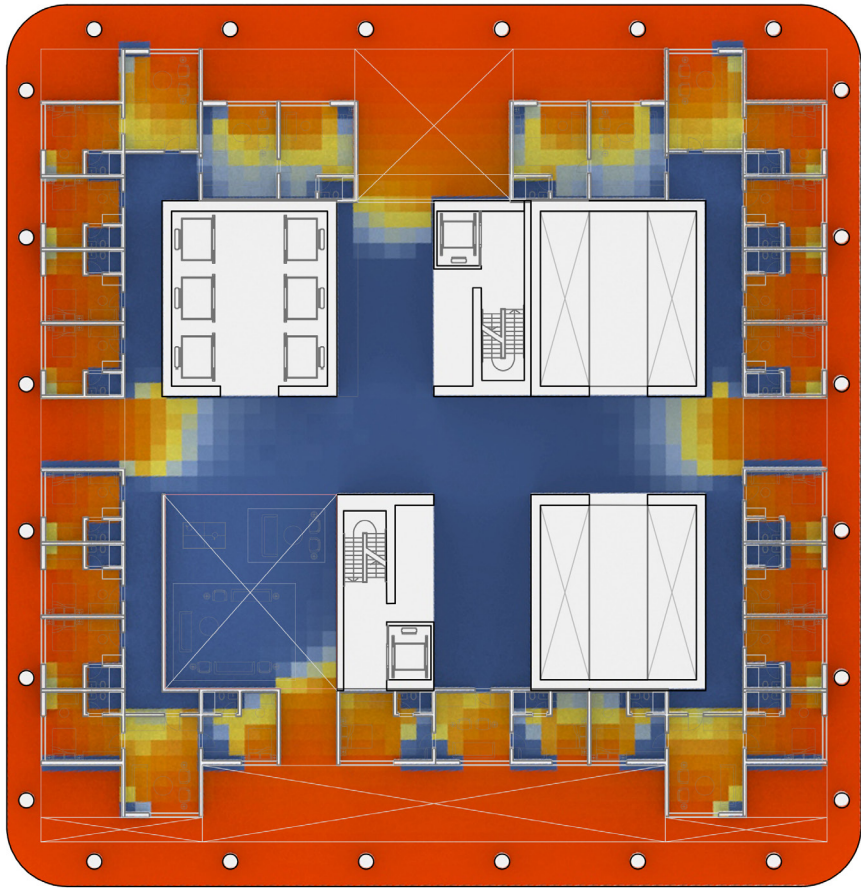


Fig 5.3.12- Daylight Autonomy- Floor 20-26 Source- Honeybee Radiance

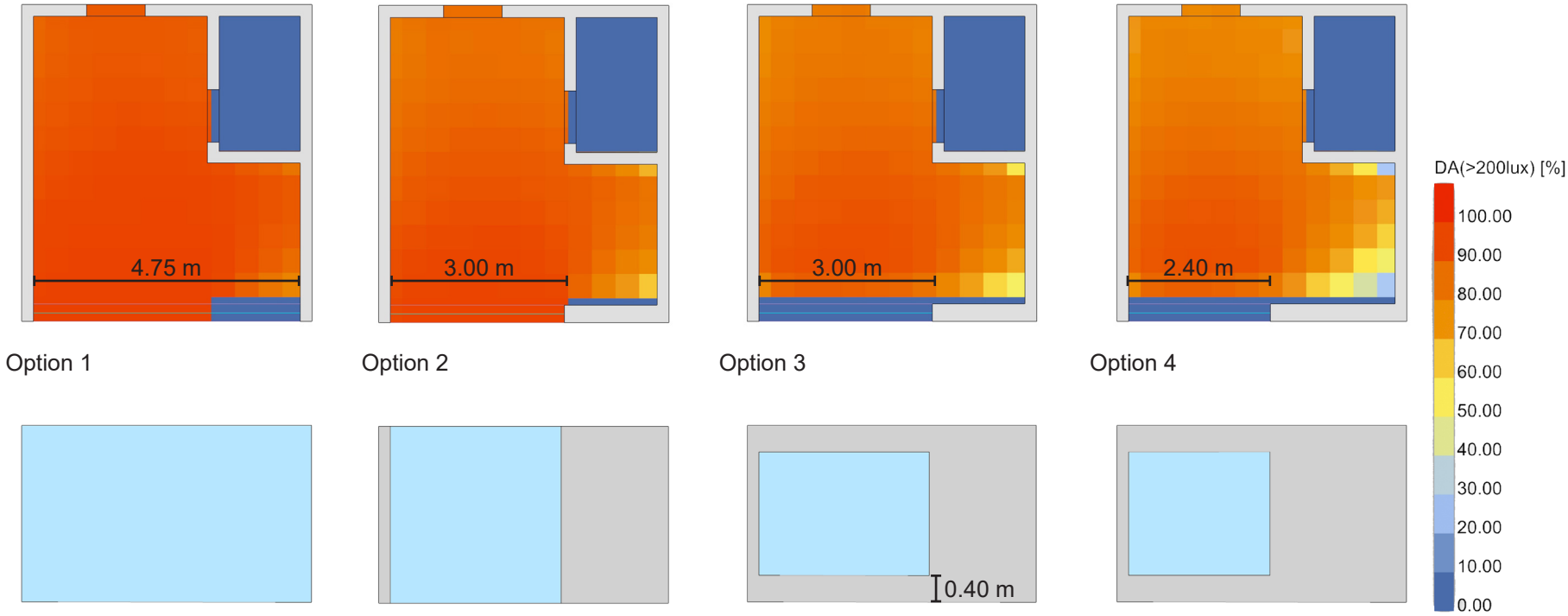


Fig 5.3.13- Daylight Autonomy- Studio Iterations

5.3.Cluster 2- Serviced Apartments

Thermal Study

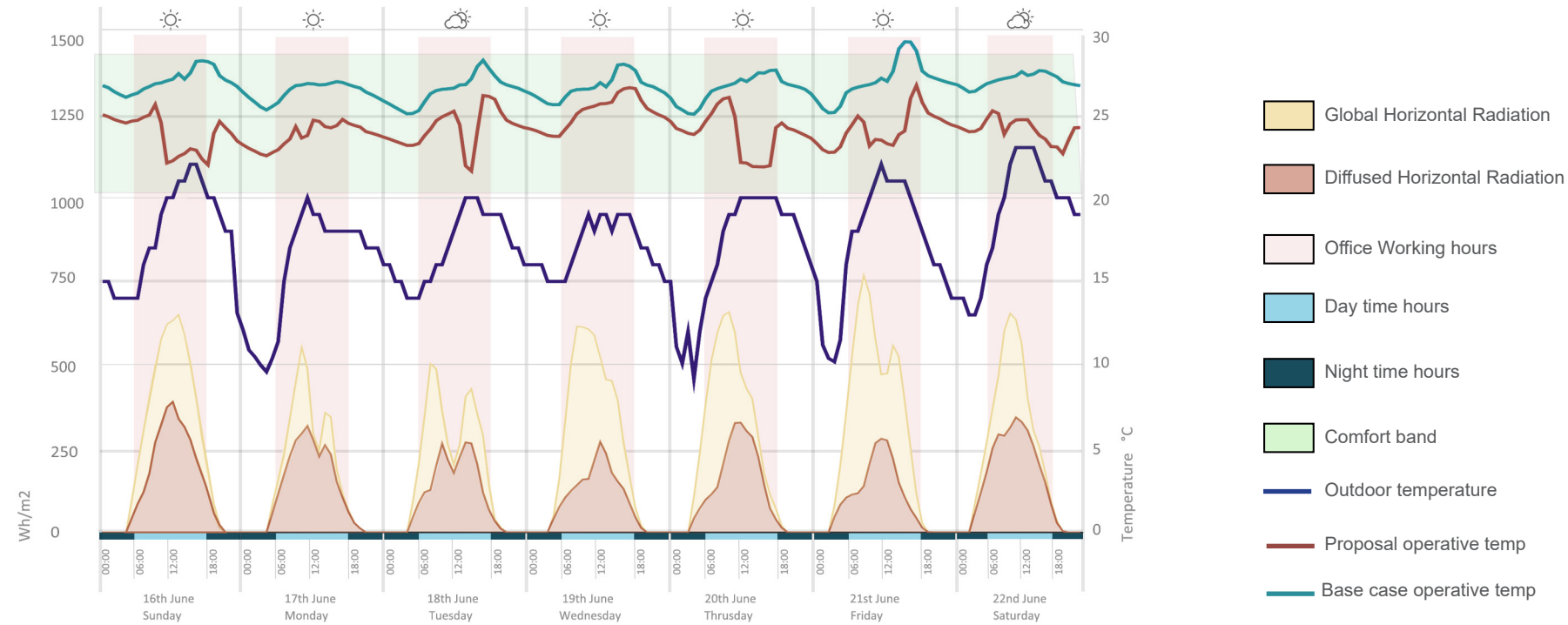


Fig 5.3.13- Thermal Studies- Studio Apartment Source- EnergyPlus

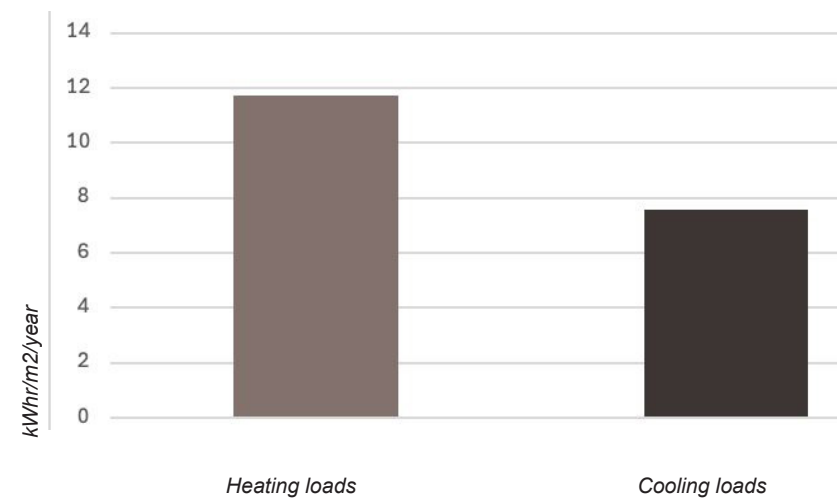


Fig 5.3.14- Thermal Studies- Base Case

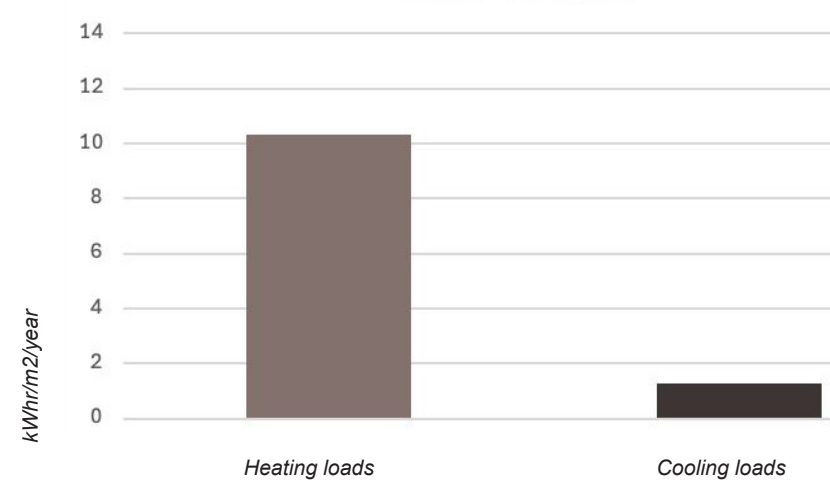


Fig 5.3.15- Thermal Studies- Proposal Design

The thermal study showcases the results for a west-facing studio apartment, highlighting how a combination of thoughtful design and key input strategies help optimize indoor comfort. The analysis reveals that the proposed design effectively keeps indoor temperatures within a comfortable range by integrating natural ventilation and shading techniques. A key feature of the design is a partially operable window (50% openable and 20% operable), allowing controlled airflow and reducing the need for mechanical cooling. Additionally, a double façade with an overhang has been introduced to manage solar heat gain, preventing overheating during the summer while still allowing for sufficient daylight.

Tests were also conducted to assess the impact of night shutters on thermal comfort. The findings suggest that night shutters have a minimal effect, indicating that the natural ventilation and shading strategies alone are enough to maintain a comfortable indoor environment. In comparison to the base case, which had a fully glazed façade and no natural ventilation, the current design significantly reduces both heating and cooling loads. The base case suffered from excessive solar heat gain and poor airflow, leading to high cooling demand. On the other hand, the new design minimizes heat buildup through better ventilation and controlled shading, creating a more stable and comfortable indoor climate.

The diagrams clearly illustrate these improvements. The thermal performance graph shows how the design keeps indoor temperatures consistent within the comfort zone, even with changing outdoor conditions. The comparison of heating and cooling loads highlights the reduction in energy demand, showcasing the efficiency of the new design. The ventilation strategy diagram further emphasizes how the operable window and double façade work together to promote cross-ventilation and regulate heat gain. Overall, the proposal enhances thermal comfort, cuts down on energy use, and creates a more sustainable and comfortable living space.

6.0 Post Design Evaluation

6.1- Intervention Results- Before vs After

The idea of restructuring and reorganizing the program for the HSBC building emerged from surveys and demographic analyses. Mixed-use buildings are today's imperative need. In a society characterized by diversity and a dynamic environment, a single-purpose private-use building is not only obsolete but also financially unsustainable. A building like HSBC – in its current state – cannot address the demands and needs of millions of people while providing high-quality performance. However, if its configuration manages to integrate a wide range of functions and uses, people would feel not only invited to use the building, but also encouraged to inhabit it. This is why the redesign proposal consists of consolidating different programs: temporary, permanent, commercial, and private. It does so not in a monolithic and uninterrupted way, but by breaking the uniformity through cuts and terraces. What was once a solid, airtight mass is now a breathable organism. What was once an icon of economic and social recognition is now a vertical village.

6.2- Total Embodied Carbon Savings

This proposal for the redesign of the HSBC building involves reusing 60% of the existing structure, resulting in notable reductions in embodied carbon.

The total savings are as follows:

- Superstructure, substructure, and finishes: 254.8 kg CO2e/m²
- Windows: 39.9 kg CO2e/m²

Moreover, the operational carbon savings are significant as well. By incorporating natural ventilation, the operational carbon is reduced to approximately 20 kg CO2e/m², due to the reduced reliance on HVAC systems, which now meet only 20% of the original demand.

6.3- Solar Radiation on a Typical Warm Season Day

There is a substantial reduction in solar exposure in the proposed design compared to the existing conditions (EC).

- North Facade: In the proposed design, 85% of the facade area experiences reduced solar radiation compared to EC. The mitigation is particularly evident in the upper and lower sections, where incident radiation values drop below 0.4 kWh/m²/day, a significant reduction from the 1.6 kWh/m²/day observed in EC.
- East Facade: The proposed design demonstrates an 80% reduction in solar radiation exposure across the east facade compared to EC. The most significant reductions occur in the lower and mid-height zones, where radiation levels fall below 1.2 kWh/m²/day, in contrast to the 3.6 kWh/m²/day recorded in EC.
- South Facade: With a 90% reduction in solar radiation, the south facade exhibits the most substantial mitigation across all orientations. The proposed design significantly decreases solar exposure, particularly in the mid and upper sections, with values reduced to below 0.8 kWh/m²/day, compared to 2.4 kWh/m²/day in EC.

West Facade: Unlike the other facades, the west facade shows a more moderate reduction, with 45% of the area experiencing lower solar radiation levels compared to EC. While radiation values decrease from 2.0 kWh/m²/day in EC to below 1.6 kWh/m²/day in the proposal, the reduction is less pronounced.

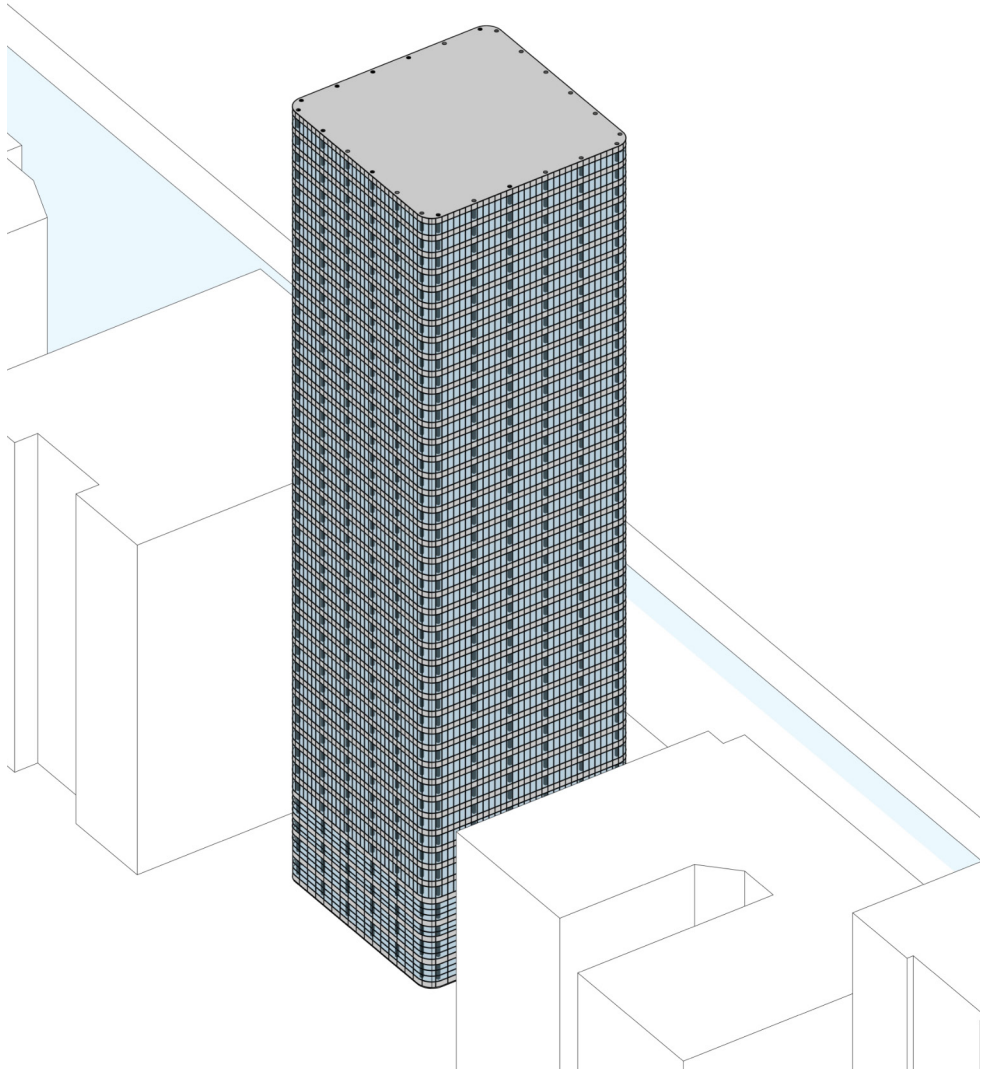


Fig 6.1.1- Existing Morphology of Building Source- Author

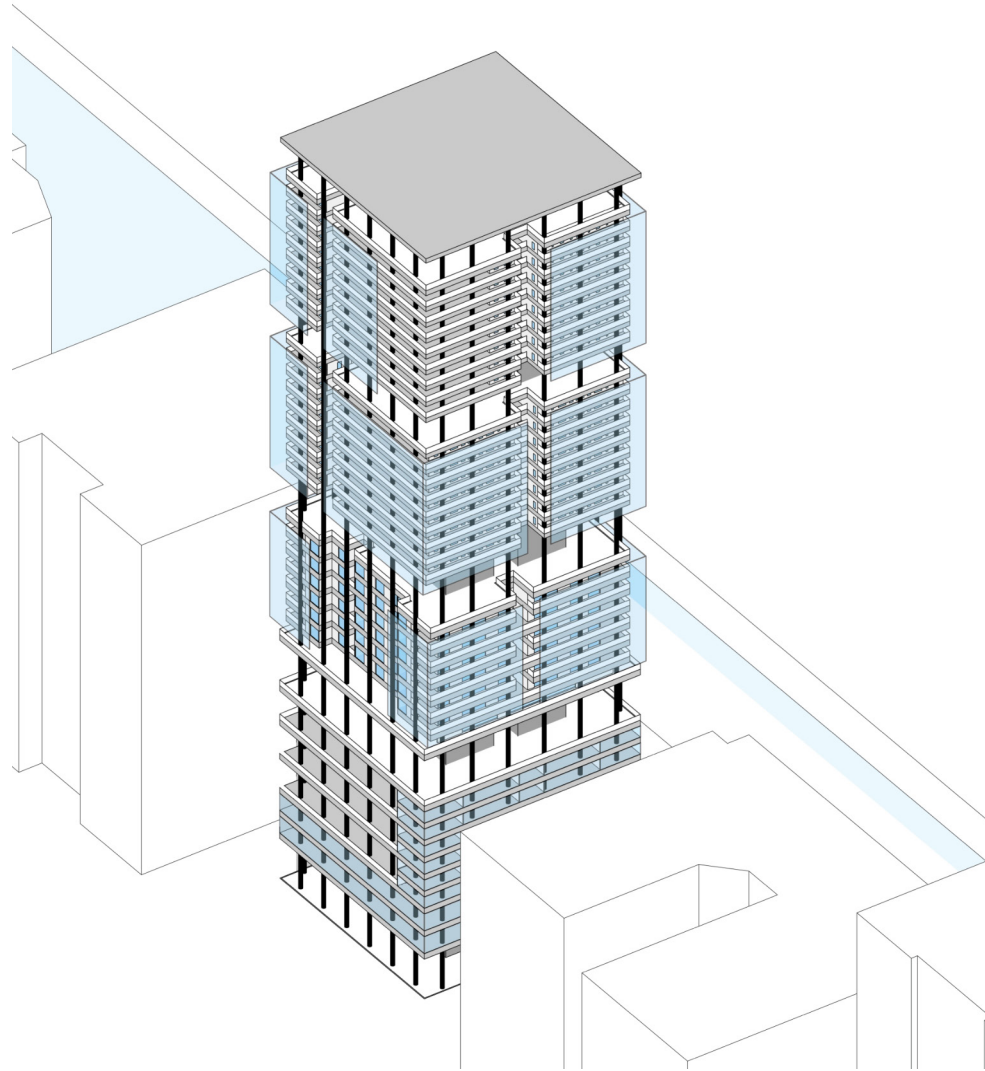


Fig 6.1.2- New Morphology of Building Source- Author

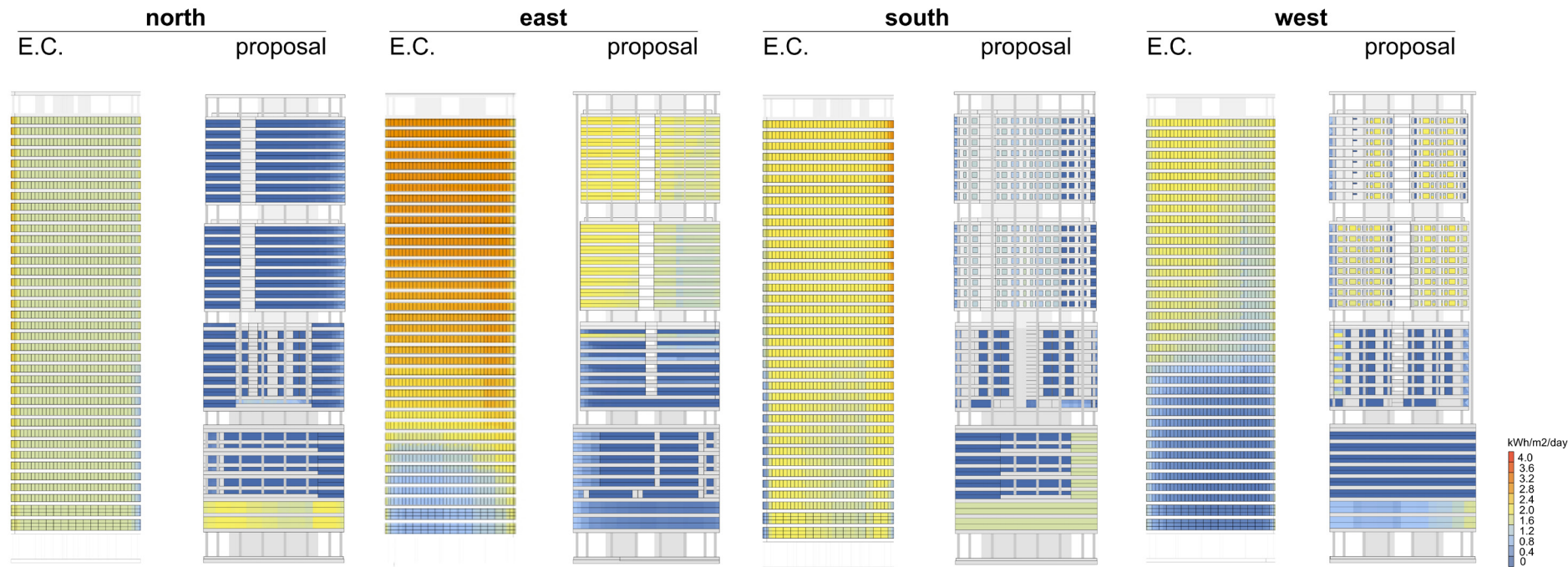


Fig 6.1.5- Solar Radiation- Existing and Proposal Comparison Source- Honeybee Radiance

6.0 Post Design Evaluation

6.4- Wind Study

Following the radiation analysis, a CFD wind analysis was conducted to assess the impact of the double-skin facade and intermediate terraces on wind pressure and ventilation. The study revealed that existing wind speeds range between 3 - 7 m/s, increasing towards the upper levels, with the south face experiencing the highest pressure due to the predominant southwest wind direction.

To address this, strategically placed terraces at regular intervals were introduced to break the wind flow and distribute pressure more evenly across the building. This design helps mitigate excessive wind loads on the south facade while promoting cross-ventilation within the structure. Additionally, high balustrades were incorporated at higher levels to shield interior spaces from direct wind impact, ensuring a more controlled indoor environment.

The voids created by terraces further optimized wind movement by channeling airflow upwards, effectively reducing incoming wind speeds while maintaining ventilation. This approach not only minimizes wind pressure but also enhances overall comfort within the building.

The double-skin facade played a crucial role in regulating wind flow, as it acts as a buffer that controls and directs airflow into interior spaces. At higher levels, it allows windows to be opened without exposing the interiors to excessive wind speeds, ensuring a balance between ventilation and wind pressure reduction.

Overall, the study confirms that the combination of terraces, balustrades, and the double-skin facade significantly reduces wind pressure and speeds, optimizing the building's form and envelope to harness positive and negative wind pressures for improved ventilation and thermal comfort. These strategies contribute to a more energy-efficient and resilient building design.

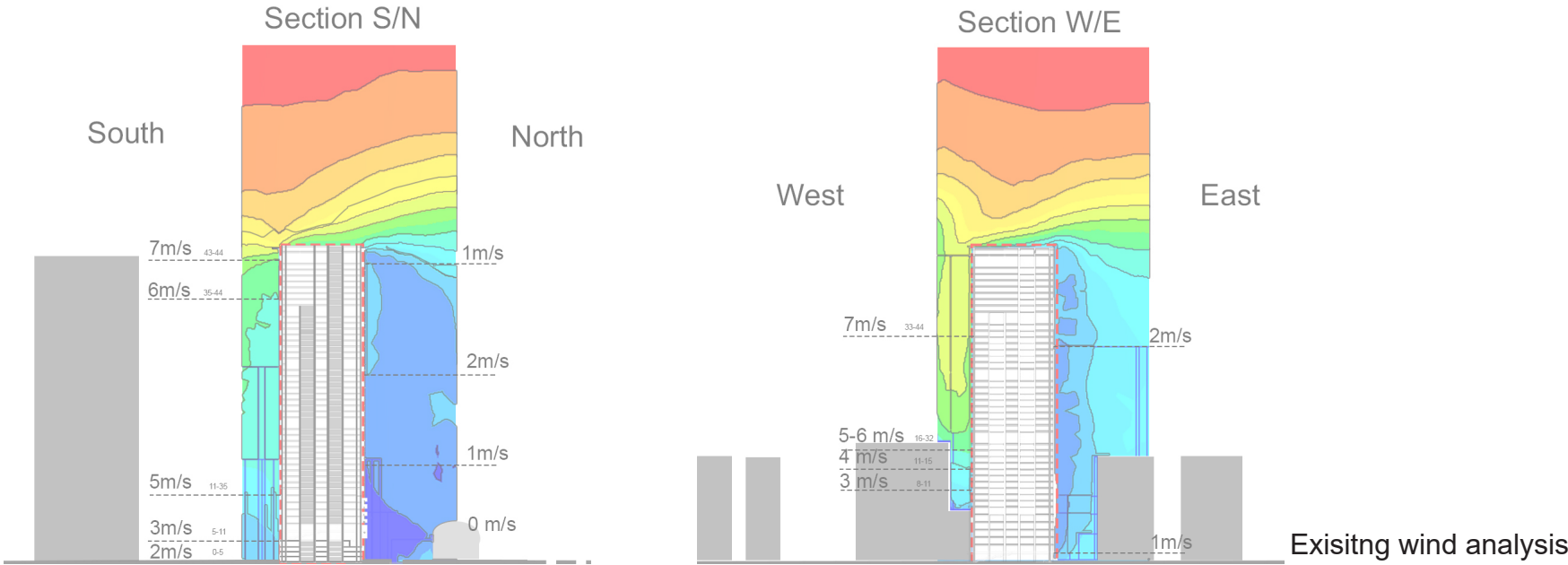


Fig 6.1.7- Window Studies- Existing Source- Autodesk CFD

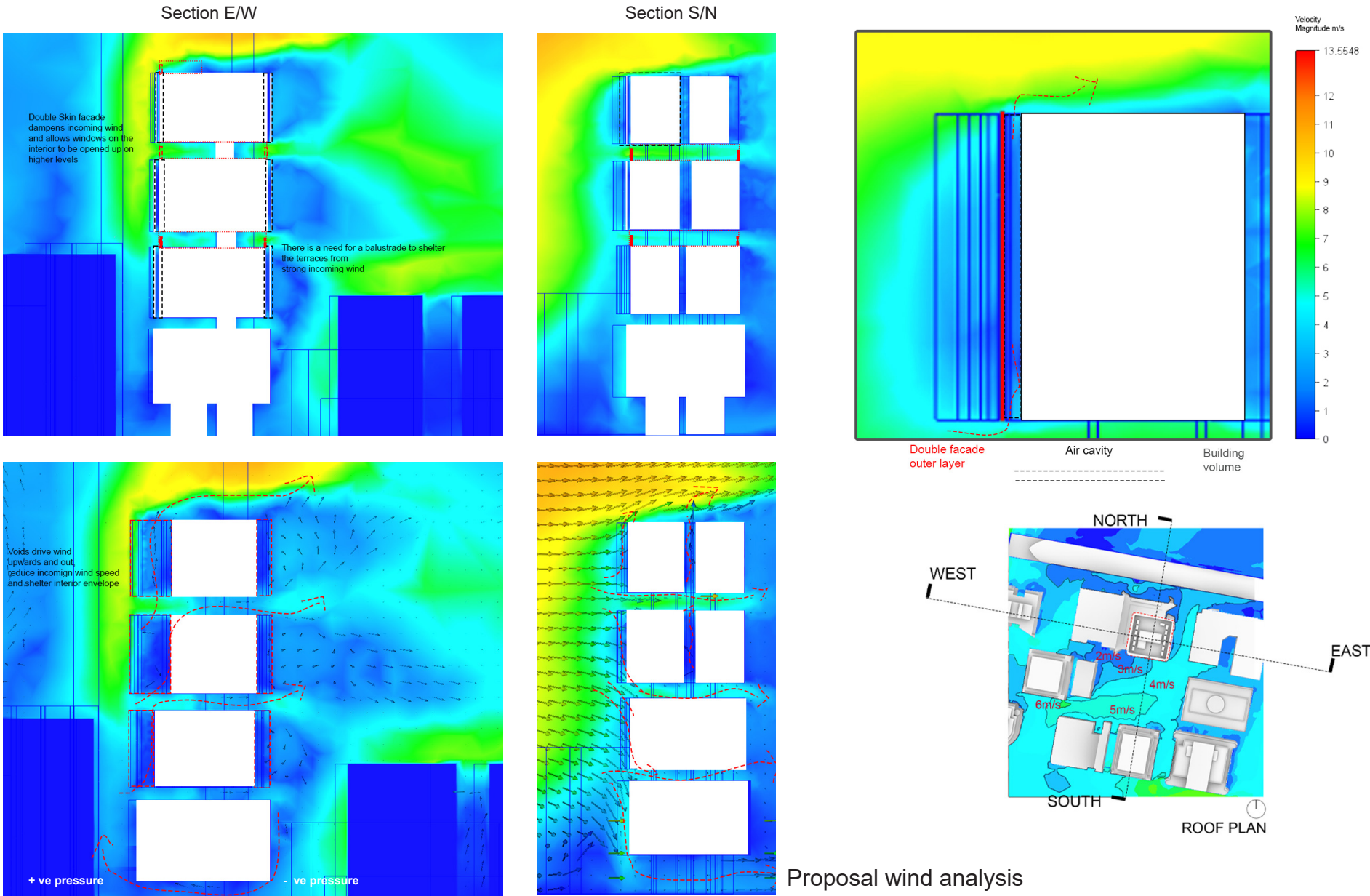


Fig 6.1.8- Window Studies- New Source- Autodesk CFD